Quantum Physics 1

Grant Talbert

May 1, 2025

BOSTON UNIVERSITY

College of Arts and Sciences
Department of Physics
PY 451

Abstract

This course was taught by Dr. Liam Fitzpatrick at Boston University during the Spring 2025 semester. We used the text Quantum Mechanics: A Paradigms Approach by David McIntyre. We also use the course website physics.bu.edu/~fitzpatr.

Contents

Chapter 1

Introduction

Lecture 1: Stern-Gerlach Experiments

Tue 21 Jan 2025 14:00

The Stern-Gerlach experiments were done in the 20s, while quantum physics was still being developed. We are given a set of silver atoms, which we will consider for this experiment to be an extremely tiny bar magnet. It has a dipole magnetic moment (vector) $\boldsymbol{\mu}$. The experiment takes a large magnet (image in notes, ill draw later) with a nonconstant magnetic field between north and south. The silver atom is then shot through the $\mathbf{B}$ field, and we look at where it comes out.

Classical mechanics says the magnetic moment $\boldsymbol{\mu}$ passing through the field $\mathbf{B}$ should deflect it as follows. Energy is given by

$$E = -\boldsymbol{\mu} \cdot \mathbf{B}.$$

We then get the force

$$\mathbf{F} = -\nabla E.$$

For example, if $\mathbf{B} = B_z(z)\hat{\mathbf{z}}$ is slightly varying along the z direction, then

$$\mathbf{F} = -\partial_z B_z \mu_z.$$

Note here that ∂_z is the partial wrt z. New notation!!!

The force then depends only on the component of the magnetic moment in the z-direction. $F \propto \boldsymbol{\mu} \cdot \hat{\mathbf{z}} = |\boldsymbol{\mu}| \cos\theta$, where θ is the angle formed with the z-axis. So

$$F_z = |\boldsymbol{\mu}| \frac{\partial B_z}{\partial z} \cos\theta.$$

Since the angle of the moments is distributed randomly, we would expect the distribution of resultant locations to be constant, since each magnetic moment corresponds to a new location, and since the angles should be distributed evenly in θ, they should be distributed evenly through the locations. However, even if they weren't *purely* random, we would get *some distribution*. It would be weird if we didn't see a distribution. So of course we don't see a distribution, because physics.

When we actually do the experiment, they only ever come out at two points, a top point and a bottom point. Since we know what $B_z(z)$ is, and since we know the mass of a silver atom, we can find the force and thus find the exact magnetic moment. We find that $\mu_z = \pm |\boldsymbol{\mu}|$ always. Even though the angles were randomly distributed, it always shows up as two possibilities.

We say it is quantized. In classical mechanics, we have a continuous distribution, but in quantum mechanics, the possibilities are all quantized into two possibilities. In quantum physics, we will talk a lot about the state of a system, and $|\Psi\rangle$ will denote the state of some system. The symbol $|\cdot\rangle$ is called a "ket".

Classically, the state is the direction of the magnetic moment, since that determines all the information we need. However, in quantum mechanics, we cannot describe the state of a system using classical physics, which is why we use kets to denote it. They are a completely new thing that doesn't show up in classical mechanics, and we will describe precisely what a ket is as we continue.

Since there are only two possibilities in the experiment, there must be only two states. We can find that those two states are

- $|+\rangle$ is the state where $\mu_z = +|\boldsymbol{\mu}|$;
- $|-\rangle$ is the state where $\mu_z = -|\boldsymbol{\mu}|$;

where again $|\boldsymbol{\mu}|$ is the moment of a silver atom.

Another thing to note is that the distribution is even throughout the possible states; 50% of the time we see $|+\rangle$, and the rest of the time we measure a state of $|-\rangle$.

Experiment two takes all the atoms that have already been deflected, and passes them through *another* magnetic field. (draw later) All the up ones keep going up, and all the down ones keep going down. Basically we didn't change anything so it stays the same. Makes sense. Ok technically we only re-measure $|+\rangle$ states.

Experiment 3 starts off the same way as experiment 1, and then proceeds *similarly* to experiment 2, but instead of measuring them in the $\hat{\mathbf{z}}$ axis again, we change the magnetic field to point in the $\hat{\mathbf{x}}$ direction. The z direction isn't special or anything, so we can do this. Classically, we would note that $\mu_z = |\boldsymbol{\mu}|$, so we would assume that $\mu_x = 0$, since the entire moment is used up in the z-component. BUT of course we get a weird answer. We once again see exactly two spots hit; exactly 50% of the time they end up in the $+x$-direction, and 50% of the time they go to the $-x$-direction. So the conclusion is the moment has multiple distinct components; orthogonal directions correspond to distinct components. We call these states $|+\rangle_x$ and $|-\rangle_x$, respectively.

Experiment 3 continues by measuring the atoms once again. So we measured them in z and found them to be distributed 50/50 in two spots. We took the $|+\rangle$ state and measured them in the x-direction and found another 50/50 distribution. However, we already measured them as $|+\rangle$ in the z-direction, so we expect them to still be in this state. This is like if we took a black sock, sorted it based on whether it was striped or not, then checked to see if it was still black. However, we *DON'T SEE THIS*. We see instead another 50/50 distribution. The particles are no longer still $|+\rangle$, but instead it's back to a completely random distribution.

So what does this mean? Measuring in z repeatedly keeps it in the same state, but when we measured it in the x component, it changed. Later, we will explain how we know this is actually what happens, and how we know this isn't the experimental machine just being shit. So measuring it in one component changed the other component. Additionally, if we only change the direction a little, rather than orthogonally, we still measure a new distribution, it's just not 50/50. So it is the act of measurement itself that changes the state of the particle. In classical physics, measuring something doesn't change it. Welcome to quantum physics, where the state $|\Psi\rangle$ cannot have a well-defined magnetic moment in two directions at the same time.

One last point, which we will discuss later. The magnetic moment $\boldsymbol{\mu}$ is related to the spin $\mathbf{S}$ by the relation

$$\boldsymbol{\mu} = \frac{ge}{2m_e}\mathbf{S}.$$

The symbol g is some constant. The spin $\mathbf{S}$ is the angular momentum of the electron when it's sitting still. Moment pointing up correspnds to

$$S_z = \frac{\hbar}{2}.$$

Definition 1.1: Kronecker Delta

The Kronecker Delta is denoted δ_{ij}, and is defined as

$$\delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}.$$

We can interpret δ_{ij} as the identity matrix $\mathbb{1}$, since it has 1s only when the indices $i = j$. Let $\mathbf{a}$ be a vector. Then $\delta_{ij} a_j = a_i$ (in Einstein notation).

Lecture 2: Linear Algebra

Thu 23 Jan 2025 14:00

The state of a system encodes all information about a system. We denote this by $|\psi\rangle$ in quantum physics. Also note that when an index is shared over two elements:

$$M_{ik} N_{kr},$$

this implies this expression really indicates a summation over the shared index:

$$M_{ik} N_{kr} := \sum_k M_{ik} N_{kr}.$$

A ket $|\psi\rangle$ represents the state of some system, but it is also a vector in a $\mathbb{C}$-vector space. This ket notation is useful since we may deal with infinite-dimensional spaces, and it also plays nicely with inner products. We always assume a vector space in quantum mechanics is a space in $\mathbb{C}$-**Vect**.

The question is, what really is $|\psi\rangle$, and how do we get information out of it? In order to do this, we need to equip our space with an inner product, making it a Hilbert space. Note that all n-dimensional $\mathbb{C}$-Hilbert spaces can be given as the matrix space $M_{n\times 1}(\mathbb{C})$. We then define $|\psi\rangle^\dagger := \pi^{-1}(\pi(|\psi\rangle)^\dagger)$, for $\pi : \mathcal{H} \to M_{n\times 1}(\mathbb{C})$ the isomorphism. More simply, given a vector

$$\begin{pmatrix} x \\ y \end{pmatrix},$$

define

$$\begin{pmatrix} x \\ y \end{pmatrix}^\dagger = \begin{pmatrix} x^* & y^* \end{pmatrix}.$$

Equivalently,

$$(z_1|v_1\rangle + z_2|v_2\rangle)^\dagger = z_1^*\langle v_1| + z_2^*\langle v_2|.$$

The Hermitian conjugate $|\psi\rangle^\dagger$ is denoted as a 'bra', $\langle\psi|$. Note that given vectors $|\phi\rangle, |\psi\rangle$, we can define an inner product

$$|\phi\rangle \cdot |\psi\rangle := \langle \phi \,|\, \psi\rangle.$$

In a more general vector space, we still denote the inner product as $\langle \phi \,|\, \psi\rangle$.

Definition 1.2: Inner Product

An inner product is a map $\langle \cdot \,|\, \cdot \rangle : \mathcal{H} \times \mathcal{H} \to \mathbb{C}$ such that

- $\langle v_1 \,|\, v_2\rangle = \langle v_2 \,|\, v_1\rangle^*$;
- $\langle v \,|\, v\rangle \geq 0$;
- $\langle v \,|\, v\rangle = 0$ iff $|v\rangle = 0$;
- $(a\langle v_1| + b\langle v_2|) \cdot (c|w_1\rangle + d|w_2\rangle) = ac\langle v_1 \,|\, w_1\rangle + ad\langle v_1 \,|\, w_2\rangle + bc\langle v_2 \,|\, w_1\rangle + bd\langle v_2 \,|\, w_2\rangle$.

Our way of pulling information out of a ket will be by taking its inner product with a state we already understand. For example, in the Stern-Gerlach experiments, we knew the state $|+\rangle$ indicated $\boldsymbol{\mu}$ pointing up, and $|-\rangle$ indicated pointing down. Assume $|+\rangle, |-\rangle, |\psi\rangle$ are all normalized with norm 1; $\langle \psi \,|\, \psi \rangle = 1$. For some reason, $|+\rangle, |-\rangle$ form a basis. We can then find the inner product of ψ with any vector using this fact. The Born Rule tells us what information this actually gives us; we will state it here, specialized to the Stern-Gerlach experiments.

Proposition 1.1

The probability of finding the state $|\psi\rangle = |+\rangle$ is

$$P_+ = |\langle + \,|\, \psi \rangle|^2 .$$

Similarly, the probability of finding the state $|\psi\rangle = |-\rangle$ is

$$P_- = |\langle - \,|\, \psi \rangle|^2 .$$

Moreover, when we measure $|\psi\rangle$ in one of these states, it turns into that state.

Since the probabilities have to sum to 1, we assume ψ has a unit norm. This follows since $\langle + \,|\, - \rangle = 0$, so the probability of finding it at both $|+\rangle$ and $|-\rangle$ is 0.

Since $|+\rangle, |-\rangle$ forms a complete basis, we can write

$$|\psi\rangle = \langle + \,|\, \psi \rangle |+\rangle + \langle - \,|\, \psi \rangle |-\rangle.$$

Then

$$\| |\psi\rangle \| = |\langle + \,|\, \psi \rangle|^2 + |\langle 0 \,|\, \psi \rangle|^2 .$$

Lecture 3: owo

Tue 28 Jan 2025 14:01

The kets $|+\rangle, |-\rangle$ form a complete basis because they are the only measurable states, and the vector space is the space of all states.

The born rule says

$$P_\pm = |\langle \pm \,|\, \psi \rangle|^2 .$$

That is, the probability of measuring a given state is the same as the norm squared of the inner product with the state vector ψ with the observable state $|\pm\rangle$.

We can also change the direction in which we measure from the $\hat{z}$ direction to the $\hat{x}$ direction. We write $|\pm\rangle_x$. We can assume WLOG that these states are normalized; that is, $\langle + \,|\, + \rangle_x = 1$. Moreover, $\langle + \,|\, - \rangle_x = 0$, because we cannot measure one state and then have it change in repeated experiments with no change. This means state vectors are orthogonal. Everything about the x-states is the same as the z-states.

It's difficult to describe how to desccibe a well-defined z and x direction at the same time; how can we show what direction $|\psi\rangle$ points in given only 2 states?

Note that if you measure $|+\rangle$ in the z-direction, then you have a 50% chance of measuring in $|+\rangle_x$ or $|-\rangle_x$, so we have that

$$P_+^{(x)} = P_-^{(x)} = \frac{1}{2}.$$

It follows that

$$|\psi\rangle_x = C_+^{(x)} |+\rangle_x + C_-^{(x)} |-\rangle.$$

In this case, we have

$$P_\pm^{(x)} = \frac{1}{2} = |\langle +_x \,|\, + \rangle|^2 = |\langle -_x \,|\, + \rangle|^2 ,$$

because measuring $|+\rangle$ gives a 1/2 chance of measuring either $+_x$ or $-_x$. Since $C_\pm^{(x)} \in \mathbb{C}$, we have that

$$\left| C_\pm^{(x)} \right| = \frac{1}{\sqrt{2}}.$$

This is because

$$C_\pm^{(x)} = \langle \pm_x \,|\, + \rangle.$$

This follows from orthogonality. Since these coefficients are complex-valued, this only tells us the probability, but we cannot discern the phase of these states. The phase of a complex number z is the θ such that $z = re^{i\theta}$. We only know the magntidude, r. This is as far as we can go experimentally, and so we can arbitrarily choose values for $C_\pm^{(x)}$ without loss of generality, so we define

$$|+\rangle = \frac{1}{\sqrt{2}}|+\rangle_x + \frac{1}{\sqrt{2}}|-\rangle_x.$$

We can use the same logic to find that the magntitude of the coefficients

$$|-\rangle = C_+^{(x)}|+\rangle + C_-^{(x)}|-\rangle$$

are also $1/\sqrt{2}$. However, we also know that since $\langle + \,|\, - \rangle$ are orthogonal, we have that

$$\langle - \,|\, + \rangle = \frac{C_+^{(x)}}{\sqrt{2}} \langle +_x \,|\, +_x \rangle + \frac{C_-^{(x)}}{\sqrt{2}} \langle -_x \,|\, -_x \rangle = 0.$$

We choose the convention that

$$C_+^{(x)} = \frac{1}{\sqrt{2}}, \qquad C_-^{(x)} = -\frac{1}{\sqrt{2}}.$$

$$\boxed{\begin{aligned} |+\rangle &= \frac{1}{\sqrt{2}}|+\rangle_x + \frac{1}{\sqrt{2}}|-\rangle_x, \\ |-\rangle &= \frac{1}{\sqrt{2}}|+\rangle_x - \frac{1}{\sqrt{2}}|-\rangle_x. \\ |+\rangle_x &= \frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle, \\ |-\rangle_x &= \frac{1}{\sqrt{2}}|+\rangle - \frac{1}{\sqrt{2}}|-\rangle. \end{aligned}}$$

So if the act of measuring outputs some $|+\rangle$, since this doesn't have a definite x-component, it's then obvious how this x-state gets scrambled. This is superposition; a great way of saying this is "Schrodinger's cat is a state in a vector space which is a linear combination of dead and alive". This is because there is no way to describe what this physicsally means in english.

Let's review the third Stern-Gerlach experiment again. We have measurements

$$|\psi\rangle \longrightarrow Z \longrightarrow X \longrightarrow Z$$

We can give the isomorphic representations

$$|+\rangle \cong \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |-\rangle \cong \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

In this same representation, we have

$$|+\rangle_x \cong \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad |-\rangle_x \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

We can also perform measurements in the Y direction, and use the same logic to determine the representations of $|\pm\rangle_y$. We have

$$|+\rangle_y = \frac{1}{\sqrt{2}}\left(|+\rangle + i|-\rangle\right),$$
$$|-\rangle_y = \frac{1}{\sqrt{2}}\left(|+\rangle - i|-\rangle\right).$$

Our representations are then

$$|+\rangle_y \cong \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix}, \qquad |-\rangle_y \cong \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -i\end{pmatrix}.$$

Now we want to talk about the generalized Born's rule. Let $\{|a_i\rangle\}_{i\in I}$ be the collection of all possible obserable states. Since we cannot observe two different states, these vectors must be orthonormal:

$$\langle a_i \mid a_j\rangle = \delta_{ij}.$$

The probability

$$P_{a_i} = \left|\langle a_i \mid \psi\rangle\right|^2.$$

Lecture 4: Operators and Measurement

Tue 04 Feb 2025 14:00

Postulate: Every observation can be represented by an operator on the space of states. For example, the measurement operators in the Stern-Gerlach experiments are S_x, S_y, and S_z. The only things you can measure are the eigenvectors of the operator of an experiment.

$$S_z = \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}.$$

Clearly this has eigenvalues $\pm\frac{\hbar}{2}$ with eigenvalues as the standard basis for $\mathbb{R}^2$. Wow, that's $|+\rangle, |-\rangle$.

$$S_x = \frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}.$$

This has the same eigenvalues with eigenvectors

$$\frac{1}{\sqrt{2}}\begin{pmatrix}1\\ 1\end{pmatrix}, \qquad -\frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -1\end{pmatrix}.$$

Wow, that's $|+\rangle_x, |-\rangle_x$! Also,

$$S_y = \frac{\hbar}{2}\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}.$$

Wow, this has the same eigenvalues but its eigenvectors are $|+\rangle_y, |-\rangle_y$. So we have thus noticed that the measurements we can observe are precisely the eigenvectors of the spin matrices.

Any hermitian operator A with eigenvalue/eigenvector pairs $a_i, |a_i\rangle$, then (assuming finite dimensionality) can be written as

$$A = \sum_{i=1}^{N} a_i |a_i\rangle\langle a_i|.$$

We can show this: node that eigenvectors with different eigenvalues are mutually orthogonal ($\langle a_i \mid a_j\rangle = \delta_{ij}$), so

$$A|a_i\rangle = \left(\sum_{j=1}^{N} a_j |a_j\rangle\langle a_j|\right)|a_i\rangle = a_i|a_i\rangle.$$

This is often the best representation for matrix operators, since we care mostly about their eigenvectors and eigenvalues. For example,

$$S_x = \frac{\hbar}{2}|+\rangle_x\langle+|_x - \frac{\hbar}{2}|-\rangle_x\langle-|_x = \frac{\hbar}{4}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix} - \frac{\hbar}{4}\begin{pmatrix}-1 & 1\\ 1 & -1\end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}.$$

What if we consider a Stern-Gerlach machine oriented along the direction $\mathbf{n}$? We can normalize it and write it in sperical coordinates:

$$\hat{\mathbf{n}} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta).$$

We are writing θ as the angle with z and ϕ as the angle in xy. Then we can write

$$\hat{\mathbf{n}}\cdot\mathbf{S} = n_x S_x + n_y S_y + n_z S_z.$$

Note that $\mathbf{S}$ is a vector of operators, so this is a largely symbolic representation.

Definition 1.3: Hermitian Operator

An operator M is *hermitian* if $M^\dagger = M$ as an operator.

In an abstract vector space, $\dagger$ is defined as the operator such that

$$\langle\phi|M^\dagger|\psi\rangle = (\langle\phi|M|\psi\rangle)^*.$$

Definition 1.4: Projection Operator

A projection operator is an operator P such that $P^2 = P$.

Examples include the operators

$$P_+ = \begin{pmatrix}1 & \\ & \end{pmatrix}, \qquad P_- = \begin{pmatrix} & \\ & 1\end{pmatrix}.$$

If we have any basis $\{|a_i\rangle\}$, then we can define

$$P_{a_i} := |a_i\rangle\langle a_i|.$$

Additionally,

$$\sum_i P_{a_i} = \mathbb{1}.$$

This is called the completeness relation.

Projection operators are how we define measurement. Suppose we perform a measurement and find the state $|a_i\rangle$. Then the final state after measurement is

$$\frac{P_{a_i}|\psi\rangle}{\sqrt{\langle\psi|P_{a_i}|\psi\rangle}}.$$

The new Born rule is then as follows. Let $\{a_i, s\}$ be the collection of all eigenvectors with eigenvalue a_i. Then

$$P_{a_i} = \sum_{s=1}^{N}|a_i, s\rangle\langle a_i, s|.$$

Lecture 5: More on Operators

Wed 05 Feb 2025 16:44

Operators represent measurements. Any hermitian operator can be represented by a sum over its eigenstates

$$A = \sum_i a_i |a_i\rangle\langle a_i|.$$

This is equivalent to saying the operator in it's eigenbasis is diagonal. Recall that a projection operator is an operator $P^2 = P$. If we were to measure an eigenvalue a_i of the observation matrix, then we know the state becomes the associated eigenstate

$$|\psi\rangle \longrightarrow |a_i\rangle.$$

We can write this as

$$|\psi\rangle \longrightarrow \tfrac{P_{a_i}|\psi\rangle}{\sqrt{\langle\psi|P_{a_i}|\psi\rangle}},$$

where

$$P_{a_i} = |a_i\rangle\langle a_i|.$$

This is just the normalized state $|a_i\rangle\langle a_i \,|\, \psi\rangle$.

Lecture 6: Statistics

Thu 06 Feb 2025 14:01

Generalized Born's Rule:

- Any obervation can be represented as a Hermitian operator acting on a space of kets;
- The possible outcomes of the measurement are the eigenvectors of the operator;
- The projector P_{a_i} of an eigenvalue a_i is given by the sum over all eigenstates $|a_i\rangle$ sharing this eigenvalue:

$$P_{a_i} = \sum_i |a_i\rangle\langle a_i|.$$

- After measurement, $|\psi\rangle$ becomes the state

$$\frac{P_{a_i}|\psi\rangle}{\sqrt{\langle\psi|P_{a_i}|\psi\rangle}}.$$

But what is probability? When we say the probability of an event occuring $\mathcal{P}_+ = |\langle + \,|\, \psi\rangle|^2$, this really involves a limit as the number of trials goes to infinity. To do this, we need to produce more $|\psi\rangle$s. You can't clone $|\psi\rangle$, so you need to redo the entire procedure that making $|\psi\rangle$ required to get a new one. Then we have

$$\lim_{N\to\infty} \frac{N_+}{N} = \mathcal{P}_+.$$

In this case, N is trials and N_+ is $|+\rangle$ outcomes.

Note that

$$\langle\psi|S_z|\psi\rangle = \frac{\hbar}{2}\left(\langle\psi\,|\,+\rangle\langle +\,|\,\psi\rangle - \langle\psi\,|\,-\rangle\langle -\,|\,\psi\rangle\right) = \frac{\hbar}{2}\left|\langle\psi\,|\,+\rangle\right|^2 - \frac{\hbar}{2}\left|\langle\psi\,|\,-\rangle\right|^2 = \frac{\hbar}{2}\left(\mathcal{P}_+ - \mathcal{P}_-\right).$$

This is called the *expectation value* of S_z in state $|\psi\rangle$.

Definition 1.5: Expectation Value

The EV of an operator X in a state $|\psi\rangle$ is

$$\langle\psi|X|\psi\rangle.$$

Classically, the EV is the sum over all probabilities of each outcome times the outcome itself. The EV of a random variable X is denoted $\langle X\rangle$. So in Quantum Mechanics, we write $\langle\psi|A|\psi\rangle$ as $\langle A\rangle$ when the state needs not be written.

I've pushed this definition forward, we need to show these definitions align. Let A be a Hermitian operator. Then

$$A = \sum_i a_i |a_i\rangle\langle a_i|.$$

Choose a state $|\psi\rangle$. Then $\mathcal{P}_{a_i} = |\langle\psi \mid a_i\rangle|^2$. The EV is also supposed to be

$$\sum_i a_i \mathcal{P}_{a_i}.$$

We have

$$\langle\psi|A|\psi\rangle = \langle\psi| \left(\sum_i a_i |a_i\rangle\langle a_i|\right) |\psi\rangle = \sum_i a_i \langle\psi \mid a_i\rangle\langle a_i \mid \psi\rangle = \sum_i a_i \left|\langle\psi \mid a_i\rangle\right|^2 = \sum_i a_i \mathcal{P}_{a_i}.$$

The variance is the standard deviation squared. We denote the standard deviation in A as ΔA. The variance

$$\Delta A = \left\langle (A - \langle A\rangle)^2 \right\rangle.$$

In quantum mechanics, we do this with respect to a state $|\psi\rangle$. So we have

$$(\Delta A)^2 = \langle\psi| \left(A - \langle\psi|A|\psi\rangle\right)^2 |\psi\rangle.$$

If c is a number, $\langle c\rangle = c$. If $\langle A\rangle$ is a number,

$$\left\langle \langle A\rangle^2 \right\rangle = \langle A\rangle^2.$$

Also, the expectation value is linear:

$$\langle cA + dB\rangle = c\langle A\rangle + d\langle B\rangle.$$

Definition 1.6: Variance

The Variance of an operator A in a state $|\psi\rangle$ is defined as

$$(\Delta A)^2 = \langle\psi|A^2 - \langle\psi|A|\psi\rangle^2|\psi\rangle.$$

$$(\Delta A)^2 = \langle\psi| \left(A - \langle\psi|A|\psi\rangle\right)^2 |\psi\rangle.$$

$$(\Delta A)^2 = \langle\psi| \langle A\rangle |\psi\rangle.$$

The fact is that you cannot get a state such that the variance in S_z and the variance in S_x are both 0. It is a mathematical impossibility. We will prove this on tuesday. However, for now, we want to know if this is true for *all* observables. When is it possible to know both at the same time?

Obviously, $|+\rangle$ in S_z and S_z^3 has 0 variance in both. So the question is really *when*.

Given an operator A, the states in which the variance is 0 is precisely the eigenvectors of A. The question is now when can you find a complete set of states that are eigenvectors of two observables A, B. You can do this *if and only if*

$$[A, B] = AB - BA = 0.$$

In this case, $[\cdot, \cdot]$ is the *commutator*.

Lecture 7: Uncertainty Principle

Tue 11 Feb 2025 14:00

Recall: given two operators A, B corresponding to two measurable quantities, we want to know when the variance in both A and B can be 0. That is, we want to find a state $|\psi\rangle$ such that the variance of A, B in $|\psi\rangle$ is 0. Recall that the variance of an operator on a state is 0 precisely when $|\psi\rangle$ is an eigenstate of A. The question "when are states with a definite A and a definite B possible" then becomes "when do two operators A and B share eigenvectors?"

Theorem 1.1

Let A, B be quantum operators. Then there exists a choice of a shared set of eigenvectors of A and B if and only if $[A, B] \coloneqq AB - BA = 0$. In this case, $[\cdot, \cdot]$ is a commutator.

Proof. Let A, B share a set of eigenvectors $\mathcal{B}$, which we can assume WLOG are orthonormal. Any operator is diagonal in its own eigenbasis, and its diagonal elements are the eigenvalues. Thus, A and B are diagonal in the eigenbasis $\mathcal{B}$. Since diagonal matrices commute, we have that $[A, B] = 0$.

Now the converse. Let $[A, B] = 0$. Let $|a\rangle$ be an eigenvector of A with eigenvalue a. There are two cases, whether $|a\rangle$ is the only eigenvector with eigenvalue a, or not.

Case 1: Consider $B|a\rangle$. By assumption, $AB = BA$. It follows that

$$A(B|a\rangle) = B(a|a\rangle) = aB|a\rangle.$$

Therefore, $B|a\rangle$ is an eigenvector of A, and since A has only one eigenvector, $B|a\rangle = b|a\rangle$. ■

For example, we can measure S_z and S_z^3 at the same time. Note that

$$[S_z, S_z^3] = S_z S_z^3 - S_z^3 S_z = 0.$$

However, we said we cannot know S_z and S_x simultaneously. Consider the commutator

$$[S_z, S_x] = \frac{\hbar^2}{4}\left(\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} - \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\right) = \frac{\hbar^2}{4}\left(\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix} - \begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}\right) = \frac{\hbar^2}{2}\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}.$$

We thus have $[S_z, S_x] = i\hbar S_y \neq 0$. If you compute all commutators, you find

$$[S_x, S_y] = i\hbar S_z, \qquad [S_y, S_z] = i\hbar S_x, \qquad [S_z, S_x] = i\hbar S_y.$$

This is a case of the more general notion of the commutation relation of angular momentum.

Uncertainty Principle: For any observables A and B, the uncertainty (in some state $|\psi\rangle$) must obey

$$(\Delta A)\,(\Delta B) \geq \frac{1}{2}\left|\langle [A, B]\rangle\right|.$$

Recall that the uncertainty ΔA is the standard deviation.

Proof. Define $\delta A \coloneqq A - \langle A \rangle$ and $\delta B \coloneqq B - \langle B \rangle$. Recall that the variance $(\Delta A)^2 = \left\langle (\delta A)^2 \right\rangle$. Since $\langle A \rangle, \langle B \rangle$ are just numbers and commute with everything,

$$[\delta A, \delta B] = [A, B].$$

Define $\mathcal{O} \coloneqq \delta A + i\lambda \delta B$ for $\lambda \in \mathbb{C}$ that will be chosen soon. Note that

$$|\mathcal{O}|\psi\rangle|^2 \geq 0$$

for all λ. Also note that since $||\psi\rangle|^2 = \langle \psi \mid \psi \rangle$, we have

$$|\mathcal{O}|\psi\rangle|^2 = \langle \psi | \mathcal{O}^\dagger \mathcal{O} | \psi \rangle = \langle \psi | \left(\delta A - i\lambda \delta B\right) \left(\delta A + i\lambda \delta B\right) |\psi\rangle.$$

Distributing, this equals

$$\left\langle (\delta A)^2 \right\rangle + \lambda^2 \left\langle (\delta B)^2 \right\rangle + i\lambda \left\langle [\delta A, \delta B] \right\rangle = (\delta A)^2 + \lambda^2 (\Delta B)^2 + i\lambda \left\langle [A, B] \right\rangle.$$

Note that $-i \langle [A, B] \rangle$ must be real. Choose $|\lambda| = \frac{\Delta A}{\Delta B}$.

We find that

$$(\Delta A)^2 + \lambda^2 (\Delta B)^2 \geq -\lambda \left| \langle [A, B] \rangle \right|.$$

It follows by algebra that

$$(\Delta B)(\Delta A) \geq \frac{1}{2} \left| \langle [A, B] \rangle \right|.$$

■

Consider the uncertainty of S_x and S_y in the state $|+\rangle$:

$$\frac{1}{2} \left| \langle [S_x, S_y] \rangle \right| = \frac{1}{2} \left| \langle i\hbar S_z \rangle \right| = \frac{1}{2} \left| i \frac{\hbar^2}{2} \right| = \frac{\hbar^2}{4}.$$

Chapter 2

Schrodinger Equation

Lecture 8: Schrodinger Equation

Thu 13 Feb 2025 14:01

The Schrodinger equation is

$$i\hbar\frac{\mathrm{d}}{\mathrm{d}t}|\Psi(t)\rangle = H(t)|\Psi(t)\rangle. \tag{2.1}$$

H is a Hermitian operator, known as the Hamiltonian, and it may or may not be time dependent. Physically, it represents the energy of $\Psi(t)$. Note that Ψ is not constant; it evolves with time. Basically the rest of this class will be spend studying equation (2.1).

2.1 Time Independent Hamiltonians

We will begin with a special case, by looking at when $H(t)$ is constant with respect to t. That is, when the Hamiltonian is independent of time:

$$\frac{\mathrm{d}H}{\mathrm{d}t} = 0.$$

First, we want to find the eigenvalues and eigenvectors of H. We will write eigenvalues and eigenvectors as $H|E_n\rangle = E_n|E_n\rangle$. The n ranges over the collection of all eigenvalues. We let $|E_n\rangle$ be an orthonormal eigenbasis of H; that is,

$$\langle E_n \mid E_m\rangle = \delta_{nm}.$$

We will now solve the Schrodinger equation, which will be shown to be basically equivalent to finding the eigenbasis $\{|E_n\rangle\}_{n\in N}$. Note that the eigenbasis for a Hermitian operator is a basis for the full space, so

1. First, find the eigenbasis $\{|E_n\rangle\}_{n\in N}$;
2. Next, take $|\Psi(t)\rangle$, and decompose in eigenbasis:

   $$|\Psi(t)\rangle = \sum_{n\in N} C_n(t)|E_n\rangle$$

   for some collection $\{C_n(t)\}_{n\in N} \subseteq \mathcal{H}$.

Plugging this into the Schrodinger equation yields

$$i\hbar\sum_{n\in N}\frac{\mathrm{d}C_n(t)}{\mathrm{d}t}|E_n\rangle = H\sum_{n\in N} C_n(t)|E_n\rangle.$$

Note that since $|E_n\rangle$ is an eigenvalue of H and H is linear, we have

$$i\hbar\sum_{n\in N}\dot{C_n}(t)|E_n\rangle = \sum_{n\in N} C_n(t)E_n|E_n\rangle.$$

In this case, since H is time-independent, the $C_n(t)$s are viewed as constants. We can now act on the left with $\langle E_m|$ for m fixed to find

$$\langle E_m| i\hbar \sum_{n\in N} \frac{\mathrm{d}C_n(t)}{\mathrm{d}t}|E_n\rangle = \langle E_m|\sum_{n\in N} C_n(t)E_n|E_n\rangle \implies i\hbar\frac{\mathrm{d}C_m(t)}{\mathrm{d}t} = E_m C_m(t)$$

because $\{E_n\}_{n\in N}$ is orthonormal. This is trivial to solve:

$$\frac{\mathrm{d}C_m(t)}{\mathrm{d}t}\frac{1}{C_m(t)} = \frac{E_m}{i\hbar} \implies \ln C_m(t) = \frac{-iE_m}{\hbar}t + C \implies C_m(t) = C_m(0)\exp\left(\frac{-iE_m}{\hbar}t\right).$$

Therefore,

$$|\Psi(t)\rangle = \sum_n C_n(0)e^{-iE_n t/\hbar}|E_n\rangle.$$

Thre is an abuse of notation to write C_n instead of $C_n(0)$; any function $f(t)$ when written as f denotes $f(0)$.

The physical signficance of the constants $C_n(t)$ changing with time indicates the probability of finding that eigenstate changes with time.

We will now do the simplest possible example. Consider $|\Psi(0)\rangle = |E_n\rangle$. Clearly we have $C_m = \delta_{mn}$, so

$$|\Psi(t)\rangle = \exp\left(\frac{-iE_n}{\hbar}t\right)|E_n\rangle.$$

Note that when you have a state multiplied by a phase $e^{i\theta}|\Psi(t)\rangle$, then $e^{i\theta}|\Psi(t)\rangle$ and $|\Psi(t)\rangle$ give identical physical predictions.

Let's measure an operator A on our $|\Psi(t)\rangle$ we found a few lines back. The only possible outcomes are the eigenstates of A, so the probability of finding $|a_i\rangle$ is

$$\mathcal{P}_{a_I} = |\langle a_i \mid \Psi(t)\rangle|^2.$$

Plugging in, we have

$$\mathcal{P}_{a_i} = \left|\langle a_i|\exp\left(\frac{-iE_n}{\hbar}t\right)|E_n\rangle\right|^2 = \left|\exp\left(\frac{-iE_n}{\hbar}t\right)\langle a_i \mid E_n\rangle\right|^2 = |\langle a_i \mid E_n\rangle|^2.$$

Since the phase always goes away, the eigenstates of a time independent Hamiltonian must not vary with time.

Now consider $\langle A\rangle$. We will show the phase goes away again. We have

$$\langle\Psi(t)|A|\Psi(t)\rangle = \exp\left(\frac{iE_n}{\hbar}t\right)\langle E_n|\exp\left(\frac{-iE_n}{\hbar}t\right)A|E_n\rangle = \langle E_n|A|E_n\rangle.$$

Note that the coefficients go away because we get the number times its complex conjugate, which is its magnitude squared, which is 1 here.

Now we will do a more complex example to make the following observation: although an entire phase infront of the state doesn't matter, differences in phases between eigenstates matter. Let

$$|\Psi(0)\rangle = C_1|E_1\rangle + C_2|E_2\rangle.$$

We have by the same logic that

$$|\Psi(t)\rangle = C_1\exp\left(\frac{-iE_1}{\hbar}t\right)|E_1\rangle + C_2\exp\left(\frac{-iE_2}{\hbar}t\right)|E_2\rangle.$$

If we measure A, we will find

$$\begin{aligned}\mathcal{P}_{a_i} &= \left|\langle a_i|\left(C_1\exp\left(\frac{-iE_1}{\hbar}t\right)|E_1\rangle + C_2\exp\left(\frac{-iE_2}{\hbar}t\right)|E_2\rangle\right)\right|^2 \\ &= \left|C_1\exp\left(\frac{-iE_1}{\hbar}t\right)\langle a_i \mid E_1\rangle + C_2\exp\left(\frac{-iE_2}{\hbar}t\right)\langle a_i \mid E_2\rangle\right|^2 \\ &= \left|\exp\left(\frac{-iE_1}{\hbar}t\right)\left(C_1\langle a_i \mid E_1\rangle + C_2\exp\left(\frac{-i(E_2-E_1)}{\hbar}t\right)\langle a_i \mid E_2\rangle\right)\right|^2 \\ &= \left|C_1\langle a_i \mid E_1\rangle + C_2\exp\left(\frac{-i(E_2-E_1)}{\hbar}\right)\langle a_i \mid E_2\rangle\right|^2.\end{aligned}$$

Note that although we can pull out a global phase, we cannot fully get rid of the phase. That means the *time dependence* of a time-independent Hamiltonian system is entirely described by the *energy differences*. This is so important that we give it a definiton.

Definition 2.1: Frequency

The frequency ω_{12} is defined as

$$\omega_{12} \coloneqq \frac{E_1 - E_2}{\hbar}.$$

Lecture 9: More on Time Independent Hamiltonians

Thu 20 Feb 2025 14:21

An example of solving the Schrodinger equation is for the case of an electron in a constant magnetic field. Let $\mathbf{B} = B_0\mathbf{z}$, and note that the magnetic moment $\boldsymbol{\mu}$ of an electron is $\frac{-e}{m_e}\mathbf{S}$, where $\mathbf{S}$ is the spin. We have $H = -\boldsymbol{\mu}\cdot\mathbf{B}$. I think $\mathbf{S}$ is a tensor, because $\mathbf{S}\cdot\mathbf{z} = S_z$. If we define $\omega_0 = \frac{eB_0}{m_e}$, we have

$$H = \omega_0 S_z.$$

Then the eigenstates are $|E_1\rangle, |E_2\rangle = |+\rangle, |-\rangle$, and the eigenvalues are $\pm\omega_0\hbar/2$. Then the solutions to the Schrodinger equation are just

$$|\psi(t)\rangle C_1 e^{-i\omega_0 t/2}|E_1\rangle + C_2 e^{i\omega_0 t/2}|E_2\rangle.$$

We can then do painful matrix calculations to find that the expectation $\langle S_z\rangle$ in this state is $\frac{\hbar}{2}\left(|C_1|^2 - |C_2|^2\right)$. We can also compute the expectation in S_x as well. Painful matrix calculations give us

$$\langle S_x\rangle = \frac{\hbar}{2}\left(C_1^* C_2 e^{\frac{it}{\hbar}(E_1-E_2)} + C_2^* C_1 e^{-\frac{it}{\hbar}(E_1-E_2)}\right).$$

Note that these two numbers are complex conjugates, so using Euler's formula, this immediately simplifies to just the cosines:

$$\langle S_x\rangle = \hbar r\cos\left(\frac{t\omega_{12}}{\hbar} + \phi\right).$$

In this case, we are representing the complex number $C_1^* C_2$ as $re^{i\phi}$. These are just initial conditions. The expectation in $\langle S_y\rangle$ is the exact same. Now consider the vector

$$\langle\mathbf{S}\rangle = \begin{pmatrix}\langle S_x\rangle(t) \\ \langle S_y\rangle(t) \\ \langle S_z\rangle(t)\end{pmatrix}.$$

Remark

We are finally calling the spin matrices the Pauli matrices. The Pauli spin matrices are the spin operators without the factor of $\hbar/2$.

$$\sigma_x = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Ok we have already unlocked MRI machines. An MRI machine needs to align the protons in your head along some $\mathbf{x}$ direction, then it switches to the σ_z Hamiltonian we just saw and watches the protons spin. Why? Good question! Let's ignore it and just learn how tf to align randomly distributed protons into a single direction. In classical physics, this means putting your brain's temperature to 10^{-44} kelvin. This would be bad, so we only want a small number of them to point up because there's a fuck ton of protons so we can still measure it even if there's only a mild deviation. The trick to doing this is to change the Hamiltonian. We want to put a small magnetic field in the perpendicular direction, as well as keeping the one in the $\mathbf{z}$ direction. We do this with a frequency, so the Hamiltonian is time dependent. Our new $\mathbf{B}$ is

$$\mathbf{B} = B_0\hat{\mathbf{z}} + B_1\cos(\omega t)\hat{\mathbf{x}} + B_1\sin(\omega t)\hat{\mathbf{y}}, \qquad B_1 \ll B_0.$$

The Hamiltonian is now

$$H = \omega_0 S_z + \frac{e}{m_e}B_1\left(\cos(\omega t)S_x + \sin(\omega t)S_y\right).$$

We call $\frac{e}{m_e}B_1 = \omega_1$. We then have

$$H = \frac{\hbar\omega_0}{2}\begin{pmatrix} 1 & \\ & -1 \end{pmatrix} + \frac{\hbar\omega_1}{2}\begin{pmatrix} & \cos(\omega t) - i\sin(\omega t) \\ \cos(\omega t) + i\sin(\omega t) & \end{pmatrix}$$

$$= \frac{\hbar}{2}\left(\omega_0\begin{pmatrix} 1 & \\ & -1 \end{pmatrix} + \omega_1\begin{pmatrix} & \exp(i\omega t) \\ \exp(i-\omega t) & \end{pmatrix}\right) = \frac{\hbar}{2}\begin{pmatrix} \omega_0 & \omega_1\exp(i\omega t) \\ \omega_1\exp(-i\omega t) & -\omega_0 \end{pmatrix}.$$

This Hamiltonian is time dependent, so we have to solve the Schrodinger equation from scratch. We are going to choose a nice basis to represent this. THe basis we like is

$$|+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

These are NOT eigenstates of H. We are just using it because it is a nice basis. Let $|\psi\rangle = C_1(t)|+\rangle + C_2|-\rangle$.

Lecture 10: Wrapping up Time Dependent Hamiltonians

Tue 25 Feb 2025 14:01

Because $|+\rangle, |-\rangle$ is a basis, we can write any state as a linear combination of $|+\rangle$ and $|-\rangle$. So let

$$|\psi\rangle = C_1|+\rangle + C_2|-\rangle.$$

Now we plug into the Schrodinger equation:

$$i\hbar\frac{\mathrm{d}}{\mathrm{d}t}|\psi(t)\rangle = H(t)|\psi(t)\rangle.$$

Using our Hamiltonian, we have

$$i\hbar\frac{\mathrm{d}}{\mathrm{d}t}|\psi(t)\rangle = \frac{\hbar}{2}\begin{pmatrix} \omega_0 & w_1\exp(i\omega t) \\ \omega\exp(-i\omega t) & -\omega_0 \end{pmatrix}\begin{pmatrix} C_1(t) \\ C_2(t) \end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix} \omega_0 C_1(t) + \omega_1\exp(-i\omega t)C_2(t) \\ \omega_1\exp(i\omega t)C_1(t) - \omega_0 C_2(t) \end{pmatrix}.$$

We can solve this differential equation but we're not going to because effort.

We will choose the initial condition $C_1 = 1$ and $C_2 = 0$, meaning that the proton(s) is(are) spin up. Solving the d2ifferential equation wrt this initial condition yields

$$|C_2(t)|^2 = \frac{\omega_1^2}{\omega_1^2 + (\Delta\omega)^2} \sin^2\left(t\frac{\sqrt{\omega^2 + (\Delta\omega)^2}}{2}\right),$$

$$|C_1(t)|^2 = 1 - |C_2(t)|^2 .$$

Here, $\Delta\omega = \omega - \omega_0$. We then have by explicit computation that

$$\langle S_z \rangle = \langle \psi(t)|S_z|\psi(t)\rangle = \frac{\hbar}{2}\left(|C_1(t)|^2 - |C_2(t)|^2\right) = \frac{\hbar}{2}\left(\frac{(\Delta\omega)^2 + \omega_1^2 \cos\left(t\sqrt{\omega_1^2 + (\Delta\omega)^2}\right)}{2}\right).$$

We have to take $\omega_1 \ll \omega_0$ in practice, since we want $\mathbf{B}_1$ to be small to avoid changing the starting point of the protons. If $\omega_1 = 0$, then the solution is just $\hbar/2$. If ω_1 is very small, then although $\langle S_z\rangle$ isn't completely constant, it is dominant. What we need is $\Delta\omega$ to be small, so the *other* term dominates. If we set $\Delta\omega = 0$, we have

$$\langle S_z \rangle = \frac{\hbar}{2}\cos(t\omega_1).$$

We want $\langle S_z\rangle = 0$ to get all of the particles to point in the x direction, so we wait for

$$t = \frac{\pi}{2\omega_1}.$$

This will set

$$\langle S_z \rangle = \frac{\hbar}{2}\cos\left(\frac{\pi}{2\omega_1}\omega_1\right) = 0.$$

This aligns all particles along the x-direction. Note that we didn't actually show this aligns them along x instead of an orthogonal direction, but it can be shown with more explicit and messier computation.

This is a special case of a more general class of common Hamiltonians. These are of the form

$$H = H_0 + V(t), \qquad H_0 = \mathrm{diag}(E_2, E_1).$$

We can write

$$H_0 = \overline{E}\begin{pmatrix} 1 & \\ & 1\end{pmatrix} + \frac{\hbar}{2}\begin{pmatrix} \omega_0 & \\ & \omega_0\end{pmatrix},$$

where $\overline{E} = \frac{E_1+E_2}{2}$ and $\hbar\omega_0 = E_2 - E_1$. In fact, it's always possible to pull out an "average" proportional to 1. Let

$$|\Psi(t)\rangle = e^{-i\overline{E}t/\hbar}|\psi(t)\rangle.$$

Since these states only differ by a phase, we haven't changed anything. However, it has an effect on the Schrodinger equation:

$$i\hbar\frac{\mathrm{d}}{\mathrm{d}t}\left(e^{-i\overline{E}t/\hbar}|\psi(t)\rangle\right) = H(t)\left(e^{-i\overline{E}t/\hbar}|\psi(t)\rangle\right).$$

We end up with

$$i\hbar\left(\frac{-i\overline{E}}{\hbar}e^{-i\overline{E}t/\hbar}|\psi(t)\rangle + e^{-i\overline{E}t/\hbar}|\psi(t)\rangle\right) = H(t)\left(e^{-i\overline{E}t/\hbar}|\psi(t)\rangle\right).$$

We can cancel out the exponentials to get

$$i\hbar\left(\frac{-i\overline{E}}{\hbar}|\psi(t)\rangle + \frac{\mathrm{d}}{\mathrm{d}t}|\psi(t)\rangle\right) = H(t)|\psi(t)\rangle.$$

Multiplying out $i\hbar$, we get

$$\overline{E}|\psi(t)\rangle + \frac{\mathrm{d}}{\mathrm{d}t}|\psi(t)\rangle = H(t)|\psi(t)\rangle.$$

We can shift the first term to the other side to obtain

$$i\hbar\frac{\mathrm{d}}{\mathrm{d}t}|\psi(t)\rangle = \left(H(t) - \overline{E}1\right)|\psi(t)\rangle.$$

We are thus always free to remove an average $\overline{H}$, as the result will remain the same. We are able to redefine our energies by a translation, as long as we shift them by the same amount. We also often see frequency reflected in Hamiltonians as well, since waves are everywhere.

Chapter 3

Entanglement

Ok now we are defining the tensor product. Given two particles with basis states $|+\rangle_1, |-\rangle_1$ and $|+\rangle_2, |-\rangle_2$, we can define a new basis in a joined Hilbert space with the tensor product:

$$|+\rangle_1 \otimes |+\rangle_2, \quad |+\rangle_1 \otimes |-\rangle_2, \quad |-\rangle_1 \otimes |+\rangle_2, \quad |-\rangle_1 \otimes |-\rangle_2.$$

Entanglement is a linear combination in this basis. For example,

$$|+\rangle_1 \otimes |+\rangle_2 + |-\rangle_1 \otimes |-\rangle_2.$$

Lecture 11: Tensors

Tue 04 Mar 2025 14:01

How do we know a superposition of states, such as $1/\sqrt{2}(|+\rangle + |-\rangle)$ is *actually* in a state lacking a well-defined z-spin? There is a famous experiment that can show an electron definitively does not have a well-defined spin, which uses entanglement. Consider two elections with spins $|\pm\rangle_1$, $|\pm\rangle_2$. The set of possible states throughout the system of 2 elections is

$$\left\{|+\rangle_1 \otimes |+\rangle_2, |+\rangle_1 \otimes |-\rangle_2, |-\rangle_1 \otimes |+\rangle_2, |-\rangle_1 \otimes |-\rangle_2\right\}.$$

This occurs in a tensor product of hilbert spaces $\mathcal{H} \otimes_{\mathbb{C}} \mathcal{H}$. A statevector can then be written as something along the lines of

$$|\psi\rangle = \frac{1}{\sqrt{2}}\left(|+\rangle_1 \otimes |+\rangle_2 + |-\rangle_1 \otimes |-\rangle_2\right).$$

A state like this is called entangled, and it can definitively show quantum mechanical things exist that do not have classical analogues. These are the Bell experiments. Today we will discuss the CHSH game, which is one of these Bell experiments. The game has two players, and each person flips a coin. They can only see the result of their own flip. After they flip their coin, they choose a color, red or blue. They want to always choose the same color, *unless* they both flip tails, in which case they want different colors. They can communicate a strategy ahead of time. The players should be far enough apart that light signals cannot pass between them during the time the game takes.

In classical mechanics, the most optimal strategy is to always pick the same color. This gives a flat 75% chance of winning the game. We can beat this using quantum entanglement. Recall our original entangled state $|\psi\rangle$. Measuring precisely one electron yields a 50% chance of measuring spin up or down. However, note that *if we measure* $|+\rangle_1$, then the state must collapse to the $|+\rangle_1 \otimes |+\rangle_2$ state. This is because there is *no way* to rewrite this state as a tensor of two sums in each electron. This implies that information has somehow traveled faster than light as well, which is extremely strange. This actually cannot transmit information however, so it's simply wavefunction collapse.

Here's how we complete the experiment. Player one measures either S_z or S_x, and player two measures $S_{\pi/8}$ or $S_{-\pi/8}$. That is, player two measures $S_{\hat{n}} = \hat{n}\cdot\mathbf{S}$, where

$$\hat{n} = \left(\pm\sin\frac{\pi}{8}, 0, \cos\frac{\pi}{8}\right).$$

Our eigenstates are

$$|+\rangle_\theta = \cos\theta|+\rangle - \sin\theta|-\rangle,$$
$$|-\rangle_\theta = -\sin\theta|+\rangle + \cos\theta|-\rangle.$$

The strategy is as follows: if player one sees heads, they measure S_z, and if they measure tails, they measure S_x. Player two measures $S_{\pi/8}$ and $S_{-\pi/8}$ if they see heads or tails, respectively. The probability of winning is

$$\cos^2\frac{\pi}{8} \approx 85.3\%.$$

Colors now correspond to the eigenvalues $\pm\hbar/2$. So if both players measure tails, they want to find different eigenvalues, and otherwise they want the same eigenvalues. Taking the probability of finding both in $+$ when measuring against the state $|+_{\theta_1}\rangle_1 \otimes |+_{\theta_2}\rangle_2$, we have

$$\mathcal{P}_{++} = |\langle\psi|\left(|+_{\theta_1}\rangle_1 \otimes |+_{\theta_2}\rangle_2\right)|^2 = \left|\frac{(\langle+|_1\otimes\langle+|_2 + \langle-|_1\otimes\langle-|_2)}{\sqrt{2}}\left(|+_{\theta_1}\rangle_1 \otimes |+_{\theta_2}\rangle_2\right)\right|^2$$

$$= \left|\frac{(\langle+|_1\otimes\langle+|_2 + \langle-|_1\otimes\langle-|_2)}{\sqrt{2}}\left(\cos\theta)1|+\rangle_1 + \sin\theta_1|+\rangle\right)1 + \cos\theta_2|+\rangle_2 + \sin\theta_2|-\rangle_2)\right|^2$$

$$= \left|\frac{1}{\sqrt{2}}\left(\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2\right)\right|^2 = \frac{1}{2}\cos^2(\theta_1-\theta_2).$$

The other ones are all the same, except $\mathcal{P}_{--}$ which is sin instead of cos. We can now plug in for θ_1, θ_2 given our setup:

$$\theta_z = 0, \qquad \theta_x = \frac{\pi}{4}, \qquad \theta_{\pi/8} = \frac{\pi}{8}.$$

Lecture 12: Entanglement Wrap Up

Thu 06 Mar 2025 14:01

This is wrong: we should have $\mathcal{P}_{++} = \mathcal{P}_{--}$, and

$$\mathcal{P}_{+-} = \mathcal{P}_{-+} = \frac{1}{2}\sin^2(\theta_1-\theta_2).$$

Suppose they get heads and heads. Then the probability of winning is equal to $\mathcal{P}_{++} + \mathcal{P}_{--}$, which is $\cos^2(\theta_1-\theta_2)$. If they both get heads, player 1 measures $\theta_1 = 0$ and player 2 measures $\theta_2 = \pi/8$, so we have

$$\mathcal{P}_{HH} = \cos^2\left(\frac{\pi}{8}\right).$$

Here, we have used the fact that cos is an even function. We can also calculate $\mathcal{P}_{TT}$.

$$\mathcal{P}_{TT} = \mathcal{P}_{+-} + \mathcal{P}_{-+} = \sin^2\left(\frac{3\pi}{8}\right) = \cos^2\left(\frac{\pi}{8}\right).$$

All other probabilities are $\cos^2(\pi/8)$ as well. This gives an overall $\cos^2(\pi/8)$ chance of winning the game.

The reason we don't see macroscopic objects in superpositions is due to quantum *decoherence*. Suppose we have a state

$$\frac{1}{\sqrt{2}}\left(|+\rangle + |-\rangle\right)\otimes|\text{Cat}\rangle.$$

At $t = 0$, the state $|A_i(0)\rangle$ is the state encompassing all information. If we evolve A_i over a very small time t, and measure $|+\rangle$, we have the state $|A_i(t)\rangle$, and if we measure $|-\rangle$, we have $|A_i'(t)\rangle$. In this case,

$$|A_i(0)\rangle \to \begin{cases} |A_i(t)\rangle & \text{happy cat} \\ |A_i'(t)\rangle & \text{kill the cat} \end{cases}.$$

Such an A_i exists for every atom in the system, so the total system is

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}\left(\bigotimes_{i=1}^{N}|A_i(t)\rangle\right) + \frac{1}{\sqrt{2}}\left(\bigotimes_{i=1}^{N}|A_i t(t)\rangle\right).$$

What is

$$\langle A_i(t) \,|\, A_i(t)\rangle$$

for some i? Since t is very small, we can approximate a Taylor expansion as $1 - \varepsilon t$. Now we can ask what the overlap is between the two components of $|\psi(t)\rangle$. We can assume ε is the same for all of them, since they are all small. This is

$$\begin{aligned} \langle \text{cat alive} \,|\, \text{cat dead}\rangle &= \prod_{i=1}^{N}\langle A_i(t) \,|\, A_i'(t)\rangle \\ &= \prod_{i=1}^{n}(1 - \varepsilon t) \\ &= (1 - \varepsilon t)^N \\ &\xrightarrow{N\to\infty} e^{-\varepsilon N t}. \end{aligned}$$

Since N is always enourmous in a classical system, $\exp(-\varepsilon N t)$ is insanely small, and overpowers ε almost instantly. This means almost instantly, the disparity between cat states goes to 0, and we only observe a single state the instant we measure. So what consitutes a measurement is a decoherence into a big system with a lot of atoms. This also explains classical probability measurements, namely that if something has spin $|+\rangle$, then it has no x spin, meaning $\langle +|S_x|+\rangle = 0$. This is because the off-diagonal elements (non orthogonal elements surrounding S_x) are killed by the massive cat states.

Chapter 4

Wavefunctions

Lecture 13: Wavefunctions

Tue 18 Mar 2025 14:01

For the next few weeks, we will simplify the math to 1-dimensional space. We will build up to the Hydrogen atom at the end of the course.

Since position is a continuous variable, we can take the state space $\mathcal{H}$ to be a function space. More precisely, $\mathcal{H} = L^2(\mathbb{C})$, and the inner product is given by

$$\langle \psi \mid \phi \rangle = \int_{\mathbb{R}} \overline{\psi} \phi \, \mathrm{d}\mu.$$

We immediately have

$$\|\psi\|^2 = \langle \psi \mid \psi \rangle = \int_{\mathbb{R}} |\psi|^2 \, \mathrm{d}\mu = \|\psi\|_2^2 .$$

It may be the case that spacetime is discretized at the planck scale or something similar, in which case the integral becomes a sum instead:

$$\textstyle\int_{\mathbb{R}} \overline{\psi} \phi \, \mathrm{d}\mu \longrightarrow \sum_{i=1}^{N} |f(x_i)|^2 \, \delta x$$

for δx the width of whatever is discretized.

We can constrain $\mathcal{H} = L^2(X)$ for X some subset of our space, and we can constrain the bounds on our norm and stuff.

We choose our basis states to be the dirac delta functions $|x_i\rangle = \delta(x - i)$. This is kind of needed, since we want to normalize our basis states. We can calculate

$$\int_{\mathbb{R}} \delta(x - i) \psi(x) \, \mathrm{d}\mu = \psi(i).$$

This follows because δ is actually a measure, so even though the function is 0 except on a set of measure 0, stuff jsut works cuz $\int \delta \, \mathrm{d}x = 1$. Therefore, we can define wavefunctions as

$$\Psi(x) = \langle x \mid \psi \rangle.$$

Recall here that $|x\rangle \coloneqq \delta_x$.

Since position is an observable, it is a hermitian operator denoted X. Since we chose a position basis $|x\rangle$ of states with definite X-value (position), they must be eigenstates of the position operator. We can thus define the position as $|\Psi(x)|^2$. The position operator in this basis is thus $\mathbb{1}$.

Since the set of delta functions is orthormal, and they are eigenvalues of the identity operator, we have a completion relation

$$\int_{\mathbb{R}} |x\rangle\langle x| \, \mathrm{d}x = 1_{\mathcal{H}}.$$

The probability of finding a particle in $[a,b]$ is

$$\int_a^b |\Psi(x)|^2 \,\mathrm{d}x.$$

Here, Ψ is the wave function $\Psi(x) = \langle x \mid \psi \rangle$.

Let H be the Hamiltonian (energy operator), with eigenstates $H|E_i\rangle = E_i|E_i\rangle$.

Lecture 14: Wavefunctions and Schrodingers Equation

Thu 20 Mar 2025 14:03

One remark I had is that the ket $|\psi\rangle$ for spin was basically extracting the spin state from the overall state $|\Psi\rangle$, and similarly the wavefunction $|\psi(x)\rangle$ is the position state which is extracted from the overall state $|\Psi\rangle$, by $\psi(x) = \langle x \mid \Psi \rangle$.

We are going to solve the Schrodinger equation for the wavefunction by solving for the eigenstates of the hamiltonian. We will be discussing hamiltonians for a single particle with position $x \in \mathbb{R}$.

Recall for a classical system, we have potential energy $V(x)$ and kinetic energy $\frac{1}{2}\frac{p^2}{m}$, so the hamiltonian is

$$H = \frac{1}{2}\frac{p^2}{m} + V(x).$$

The potential is related to a force applied to the system by

$$\frac{\mathrm{d}V(x)}{\mathrm{d}x} = -F.$$

We will be studying potentials over the next few classes, and will start with a particle in a box. We will be studying time-independent potentials, because it's easier, and since many interesting potentials are time-independent.

To make this quantum, we need to make p an operator. Since p is momentum, and thus an observable, we need it to be an operator. Similarly, we need $V(x)$ to be an operator. Since $V(x)$ is an operator, we can just do $V(X)$, where X is the position operator. This is simple, since $V(X)|x\rangle = V(x)|x\rangle$, since $|x\rangle$ is an eigenstate of X with eigenvalue x. We can thus evaluate $V(X)|\psi\rangle$ by writing $|\psi\rangle$ as a linear combination in $|x\rangle$:

$$|\psi\rangle = \int \psi(x)|x\rangle \,\mathrm{d}x.$$

For example,

$$V(x) = x^2.$$

We then have

$$V(X)|\psi\rangle = X^2|\psi\rangle = X^2 \int \psi(x)|x\rangle \,\mathrm{d}x = x^2|\psi\rangle.$$

So for example,

$$\langle\psi|X^2|\psi\rangle = \int x^2 |\psi(x)|^2 \,\mathrm{d}x.$$

Now the question is what to define p as. We define

$$P \cong i\hbar\frac{\mathrm{d}}{\mathrm{d}x}.$$

Evaluating $\langle\psi|P|\psi\rangle$ goes like this.

$$\langle\psi|P|\psi\rangle = \int \psi^*(x)\left(-i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\right)\psi(x)\,\mathrm{d}x.$$

We then have

$$\langle x|P|\psi\rangle = -i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\psi(x).$$

This is simply a definition. Since quantum is more fundamental than classical, we don't "derive" momentum as an operator, but rather we would show that this momentum operator behaves like classical momentum in the limit as quantum approaches classical.

We thus have

$$\begin{aligned}\langle x|H|\psi\rangle &= H\psi(x)\\ &= \left(\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x)\right)\psi(x)\\ &= \frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}\psi(x) + V(x)\psi(x).\end{aligned}$$

By the Schrodinger equation, we have

$$\begin{aligned}-i\hbar\frac{\mathrm{d}}{\mathrm{d}t}|\psi\rangle &= H|\psi\rangle\\ &= \left(\frac{1}{2m}P^2 + V(X)\right)|\psi\rangle\\ &= \left(\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(X)\right)|\psi\rangle\\ &= \frac{\hbar^2}{2m}\psi(x) + V(x)\psi(x).\end{aligned}$$

To solve the Schrodinger equation, we want to solve the eigenvalue problem for this H. This is a very deep fact. Since H represents energy, and we only measure eigenstates, this means the only observable energy values E satisfy

$$\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}\psi(x) + V(x)\psi(x) = E\psi(x).$$

Now we will prove the pop culture version of the uncertainty principle. This is the canonical commutation relation

$$[X,P] = i\hbar.$$

We have

$$\begin{aligned}\langle\psi_1|[X,P]|\psi_2\rangle &= \langle\psi_1|XP - PX|\psi_2\rangle\\ &= \langle\psi_1|XP|\psi_2\rangle - \langle\psi_1|PX|\psi_2\rangle\\ &= \int \psi_1^*(x)x\left(-i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\right)\psi_2(x)\,\mathrm{d}x - \int \psi_1^*(x)\left(-i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\right)x\psi_2(x)\,\mathrm{d}x\\ &= \int \psi_1^*(x)x\left(-i\hbar\frac{\mathrm{d}\psi_2(x)}{\mathrm{d}x}\right) - \psi_1^*(x)\left(-i\hbar\right)\left(\frac{\psi_2(x)\mathrm{d}x}{\mathrm{d}x} + x\frac{\mathrm{d}\psi_2}{\mathrm{d}x}\right)\mathrm{d}x\\ &= \text{stuff cancels}\\ &= \int \psi_1^*(x)i\hbar\psi_2(x)\,\mathrm{d}x\\ &= i\hbar\langle\psi_1\,|\,\psi_2\rangle.\end{aligned}$$

Using the uncertainty principle, we have

$$\Delta X\Delta P \geq \frac{1}{2}\left\langle|i\hbar|\right\rangle = \frac{\hbar}{2}.$$

Using this, we can show

$$\langle P \rangle = m \frac{\mathrm{d}}{\mathrm{d}t} \langle X \rangle .$$

This implies the classical definition

$$p = m \frac{\mathrm{d}x}{\mathrm{d}t}.$$

Lecture 15: More Schrodinger Equation

Tue 25 Mar 2025 14:04

Ok they are actually partial derivatives because $\psi(x,t)$ since ψ has time dependence. We usually write it as a total derivative though, as we usually write the state as $|\psi, t\rangle$ for some reason.

To show

$$\langle P \rangle = m \frac{\mathrm{d}}{\mathrm{d}t} \langle X \rangle ,$$

we write

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle \psi | X | \psi \rangle = \frac{\mathrm{d}}{\mathrm{d}t} \langle \psi | \int x |x\rangle\langle x| \,\mathrm{d}x |\psi\rangle = \frac{\mathrm{d}}{\mathrm{d}t} \int \langle \psi \mid x \rangle x \langle x \mid \psi \rangle \,\mathrm{d}x = \frac{\mathrm{d}}{\mathrm{d}t} \int \psi^*(x) x \psi(x) \,\mathrm{d}x.$$

We write this as

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle \psi, t | X | \psi, t \rangle.$$

The time dependence of states comes from the Schrodinger equation

$$i\hbar \frac{\mathrm{d}}{\mathrm{d}t} |\psi, t\rangle = H |\psi, t\rangle = \frac{P^2}{2m} |\psi, t\rangle + V(x) |\psi, t\rangle.$$

Taking the Hermitian conjugate gives us

$$i\hbar \frac{\mathrm{d}}{\mathrm{d}t} \langle \psi, t| = - \langle \psi, t| H,$$

because H is Hermitian. Using our above equation, we have

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle \psi, t | X | \psi, t \rangle = \left(\frac{\mathrm{d}}{\mathrm{d}t} \langle \psi, t| \right) X |\psi, t\rangle + \langle \psi, t | X \left(\frac{\mathrm{d}}{\mathrm{d}t} |\psi, t\rangle \right).$$

Plugging this in, we obtain

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle X \rangle = \frac{1}{i\hbar} \left(- \langle \psi, t | H X | \psi, t \rangle + \langle \psi, t | X H | \psi, t \rangle \right) = - \frac{1}{i\hbar} \langle \psi, t | \left(-HX + XH \right) |\psi, t\rangle = \frac{1}{i\hbar} \langle [X, H] \rangle .$$

This generalizes to any *time-independent* operator X, not just position. This shows us that

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle A \rangle = \frac{1}{i\hbar} \langle [A, H] \rangle .$$

Note that $[A, BC] = [A, B] \, C + B \, [A, C]$.

$$[X, H] = \left[X, \frac{p^2}{2m} + V(x) \right] = \left[X, p^2/2m \right] + \left[X, V(x) \right].$$

Recall that $[X, p] = i\hbar$. We have

$$\frac{1}{2m} \left([X, p] \, p + p \, [X, p] \right) = \frac{1}{2m} \left(i\hbar p + i\hbar p \right) = \frac{i\hbar}{m} p.$$

Additionally, $[X, V(x)] = 0$, since they share eigenstates. We can plug this back in to obtain

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle X \rangle = \frac{1}{i\hbar} \langle [X, H] \rangle = \frac{1}{i\hbar} \frac{i\hbar}{m} \langle p \rangle = \frac{1}{m} \langle p \rangle .$$

Therefore, $m \frac{\mathrm{d}X}{\mathrm{d}t} = P$. This satisfies the classical relation for position and momentum, so we have that the momentum operator behaves as expected.

4.1 Solving Schrodinger's Equation

We will take various potentials $V(x)$, and solve the Schrodinger equation in that potential. The simplest one is in fact the infinite square well:

$$V(x) = \begin{cases} 0, & 0 < x < L \\ \infty, & \text{otherwise} \end{cases}.$$

We want to find all eigenvals/vecs of H in this potential:

$$H = \frac{p^2}{2m} + V(x).$$

We then have the eigenvalue problem

$$H|\psi\rangle \cong \left(-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x)\right)\psi(x) = E|\psi\rangle.$$

Since V goes to infinity outside of $(0, L)$, the wavefunction must be 0 in this interval, since the particle cannot go there. We can then impose boundary conditions on our equation. We will start by solving it in the region where $V(x) = 0$:

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} = E\psi(x) \implies \frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} = -\frac{2mE}{\hbar^2}\psi(x).$$

Set $k = \sqrt{\frac{2mE}{\hbar^2}}$. We then have

$$\psi(x) = \exp\left(\pm ikx\right).$$

Our general solution is thus

$$\psi(x) = A'e^{ikx} + B'e^{-ikx}.$$

To identify E from k, we impose the boundary conditions. We need to find the values of k such that we oscillate precisely an integer number of times within the well, since otherwise it would be nonzero on the boundary (the wavefunction needs to be continuous).

Our boundary conditions are that $\psi(0) = \psi(L) = 0$. Plugging this in, we obtain

$$0 = A'e^0 + B'e^0 = A' + B'.$$

Therefore, $A' = -B'$. Plugging in L, we have

$$0 = A'e^{ikL} + B'e^{-ikL} = A'e^{ikL} - A'e^{-ikL}.$$

This forces $e^{ikL} - e^{-ikL} = 0$, and thus

$$\cos kL + i\sin kL - \cos(-kL) - i\sin(-kL) = 2i\sin kL.$$

This quantizes k immediately, as $\psi(x) \neq 0$ since the particle exists, and thus

$$kL \in \pi\mathbb{Z} \implies \sqrt{\frac{2mE}{\hbar^2}}L = \pi n \implies E = \frac{n^2}{L^2}\frac{\hbar^2}{2m}\pi^2 = \frac{n^2\hbar^2\pi^2}{2Lm}$$

for any $n \in \mathbb{Z}^+$. We cannot have $n < 0$ for obvious reasons. Therefore, after normalization we have

$$\psi(x) = A\sin\left(\frac{n\pi x}{L}\right) = A\sin\left(k_n x\right), \qquad k_n := \frac{n\pi}{L}.$$

The sin instead of the exponential comes from the decomposition into sin we did earlier, but more general. The normalization $A' \to A$ is given by

$$1 = \langle \psi \mid \psi \rangle = \int_{\mathbb{R}} |\psi(x)|^2 \, \mathrm{d}x \implies \langle \psi_n \mid \psi_n \rangle = \int_0^L |A \sin(k_n x)|^2 \, \mathrm{d}x = |A|^2 \int_0^L \sin^2(k_n x) \, \mathrm{d}x.$$

This integral evaluates to $L/2$, giving the normalization factor for A. We then have $A = \sqrt{2/L}$, and each $|\psi_n\rangle$ has norm 1:

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin(k_n x).$$

These are the solutions to the Schrodinger equation. Since the energy levels grow like $E_n \propto n^2$, we have $E_n = n^2 E_1$. Since these functions are all symmetric, the probability in the left/right halves of the box are 1/2.

Lecture 16: Solving Another Schrodinger Equation

Thu 27 Mar 2025 14:00

In position space, this means there exist $\{c_n\} \subseteq \mathbb{C}$ such that

$$\psi(x) = \sum_n c_n \psi_n(x) = \sum_n c_n \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi}{L} x\right)$$

for any wavefunction ψ (subject to this potential). This is a Fourier series because of course. One more thing to say is that

$$\sum_n |E_n\rangle\langle E_n| = \mathbb{1},$$

and since $|E_n\rangle$ in position space is ψ_n, we have the statement.

Now we will do something that can actually happen. We will do the finite square well

$$V(x) = \begin{cases} 0, & -\frac{L}{2} < x < \frac{L}{2} \\ V_0, & \text{otherwise} \end{cases}.$$

We will call $L/2 =: a$, and thus $L = 2a$. The boundary condition is the hardest part to figure out. We will start by taking our solution for $x \in [-a, a]$, which was

$$\psi(x) = Ce^{ikx} + De^{-ikx}.$$

Recall that

$$-\frac{2mE}{\hbar^2} = k^2.$$

For $V(x) \neq 0$, we have the equation

$$-\frac{\hbar^2}{2m} \frac{\mathrm{d}^2}{\mathrm{d}x^2} \psi(x) = (E - V_0)\psi(x).$$

We then have

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2} \psi(x) = \frac{(V_0 - E)2m}{\hbar^2}.$$

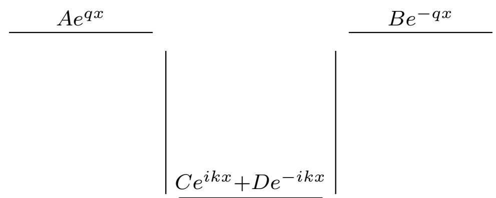
Define

$$q^2 := \frac{2m}{\hbar^2}(V_0 - E).$$

We then have

$$\psi(x) = Ae^{qx} + Be^{-qx}.$$

If $E < V_0$, then classically the particle wouldn't be able to leave the potential well. Note that this linear combination has to be different on both sides of the well; if the Ae^{qx} term were nonzero on the right side, then it would go to infinity and be non-normalizable. We then have to take precisely one solution per side of the well.



The thing to note is that these wavefunctions have a nonzero probability of leaving the box, even though their energy is not high enough. This is called quantum tunneling. However, note that the exponentials go down very quickly, so the probability is fairly small. This is non-relativistic physics, so we will ignore the possibility of going faster than light. This is reconsiled later with relativistic mechanics, which are way harder.

The final step is to impose boundary conditions on our solutions. One of these conditions is that ψ be continuous, and moreover the first derivative should be continuous, which can be shown with some manipulation and the Fundamental Theorem of Calculus. This doesn't hold for discontinuous potentials in general, however.

Imposing continuity gives us

$$Ae^{-qa} = Ce^{-ika} + De^{ika}, \qquad Be^{-qa} = Ce^{ika} + De^{-ika}.$$

Convering into sins and cosines, $\psi(x) = C\sin kx + D\cos kx$, we have

$$k\frac{D\sin ka + C\cos ka}{D\cos ka - C\sin ka} = q = k\frac{D\sin ka - C\cos ka}{D\cos ka + C\sin ka}.$$

Simplifying shows us that $CD = 0$, so either C or $D = 0$. This gives us even solutions $D\cos kx$, and odd solutions $C\sin kx$. This gives us $q = k\tan ka$ if $C = 0$, and $q = -k\cot ka$ for $D = 0$. Our boundary conditions we already imposed were that as $x \to \pm\infty$, $\psi(x) \to 0$. Since k and q are not independent:

$$k^2 = \frac{2mE}{\hbar^2}, \qquad q^2 = \frac{2m}{\hbar^2}(V_0 - E),$$

we have

$$q^2 = \frac{2m}{\hbar^2}V_0 - k^2,$$

and thus for even solutions

$$\sqrt{\frac{2m}{\hbar^2}(V_0 - E)} = \sqrt{\frac{2mE}{\hbar^2}}\tan\left(a\sqrt{\frac{2mE}{\hbar^2}}\right).$$

This is impossible to solve in general, but intuitively we know there will be finitely many solutions. After E_n is high enough for the particle to leave the box, it will just leave the box.

Lecture 17: Unbound States

Tue 01 Apr 2025 14:00

The question is now how to deal with the unbound states. The simplest possible state has $V(x) = 0$, so the eigenvalue equation becomes

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}\psi(x) = E\psi(x).$$

Setting

$$k \coloneqq \sqrt{\frac{2mE}{\hbar}},$$

we find instantly that the solutions are $\exp(\pm ikx)$. We have

$$\psi(x) = Ae^{ikx} + Be^{-ikx}.$$

Our boundaries are now $\psi \to 0$ as $x \to \infty$. However, the magnitudes of exponentials are always 1, so the magnitude here will remain constant for all x, and cannot go to infinity. Writing the solutions in Euler's formula, we have

$$A\cos(kx) + iA\sin kx, \qquad B\cos kx - iB\sin kx.$$

So we cannot in fact impose boundary conditions, since they just continuously oscillate. This allows any A or B to solve the equation, since there need not be bounds. We just require the wavefunction me normalized. We thus have translation symmetry as the origin is no longer important, so it's kinda like Affine space.

Let's consider the difference between these two states. We will consider the action of the momentum operator on ψ_+, ψ_-. Note that $[H, P] = 0$ in this case, so we can measure P without changing the state. So let's find the eigenvalues of P, which will also be eigenvalues of H. That is, we want

$$P|p\rangle = p|p\rangle.$$

This is solved by exponentials:

$$\frac{\mathrm{d}}{\mathrm{d}x}e^{kx} = \frac{i\hbar}{p}e^{(i\hbar/p)x}.$$

Let $\psi_p(x) = A\exp(i\hbar x/p)$ for A some normalization constant. Note that since $\psi(x) = \langle x \,|\, \psi\rangle$, we have that $\psi_p(x) = \langle x \,|\, p\rangle$. This is a fundamental fact that $\langle x \,|\, p\rangle$ is an exponential.

Going back to our solutions

$$\psi(x) = Ae^{ikx} + Be^{-ikx},$$

we find that their momentums are $\hbar k$ and $-\hbar k$, respectively. The symmetry between the solutions is that they have opposite momenta. If we view the real part of an exponential as a wave, which has wavelength $\lambda = \frac{2\pi}{k} = \frac{2\pi\hbar}{p}$, this is called the De Broglie wavelength. This shows that for any free particle, the wavefunction behaves wavelike - and thus so does the probability distribution (i think). It has a wavelength $\frac{2\pi\hbar}{p}$.

Now let's consider the normalization constant. Consider

$$\langle p \,|\, p'\rangle = A_p A_{p'} \int_{-\infty}^{\infty} \psi_p^*(x)\psi_{p'}(x)\,\mathrm{d}x = A_p A_{p'} \int_{-\infty}^{\infty} e^{i(p-p')x/\hbar}\,\mathrm{d}x.$$

This integral is 0 (analytic) unless $p - p' = 0$, when it goes to infinity. Doing stuff with power series and Fourier transforms shows that this integral is

$$A_p A_{p'} 2\pi\hbar\delta(p - p').$$

The $\hbar$ outside came from a change of variables. More generally,

$$\int_{-\infty}^{\infty} e^{ikx}\,\mathrm{d}x = 2\pi\delta(k).$$

This makes sense since these are eigenvectors. To normalize this, we are going to choose $\langle p \,|\, p'\rangle = \delta(p - p')$, and thus we set

$$A_p \coloneqq \frac{1}{\sqrt{2\pi\hbar}}.$$

Our momentum eigenfunctions are thus

$$\langle x \,|\, p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar} = \psi_p(x).$$

Our next problem concerns the time evolution of these states. Let $|\psi, t=0\rangle = |\psi_p\rangle$; that is, $\psi(x,0) = \psi_p(x)$. Using Schrodinger's equation, we have

$$i\hbar \frac{\partial}{\partial t}\psi(x,t) = E_p \psi(x,0).$$

Since this is already an eigenvalue, the solution is an exponential in time:

$$\frac{\partial \psi(x,t)}{\partial t} = \frac{E_p}{i\hbar}\psi(x,0) \implies \psi(x,t) = e^{-iE_p t/\hbar}\psi_p(x,0).$$

Plugging in, we have

$$\psi(x,t) = \frac{1}{\sqrt{2\pi\hbar}} e^{-iE_p t/\hbar} e^{ipx/\hbar} = \frac{1}{\sqrt{2}} \exp\left(ip(x - v_p^{(ph)} t)\right),$$

where $v_p^{(ph)} = E_p/p = \frac{p^2}{2mp} = \frac{p}{2m}$. This is called the phase velocity. However, since $p = mv$, we would expect the velocity be p/m, not $p/2m$. This is called the *phase velocity* because it's *not* the velocity, but rather the speed at which the phase changes. Since the particle is spreading out according to the Schrodinger equation, and it is spread out over all space (since it is in an eigenstate), it can't really be moving. This can be brought back to the canonical commutation relation

$$\Delta x \Delta p \geq \frac{\hbar}{2}.$$

If we know p with 100% precision, then

$$\Delta x \geq \frac{\hbar}{2}\frac{1}{0}.$$

This makes no sense. That is, Δx is unknowable. However, nothing is actually spread through out ALL space, so particles actually look like wave *packets*, which are wavy gaussians. It would then have a fairly well-defined position and momentum. There is then a well-defined notion of velocity, since we can consider the speed of the bump in the particle.

Recall the decomposition

$$|\psi\rangle = \int_{-\infty}^{\infty} \psi(x)|x\rangle \, \mathrm{d}x.$$

We can also decompose into momentum space:

$$|\psi\rangle = \mathbb{1}|\psi\rangle = \int_{-\infty}^{\infty} |p\rangle\langle p \,|\, \psi\rangle \, \mathrm{d}p = \int_{-\infty}^{\infty} \psi(p)|p\rangle \, \mathrm{d}p.$$

The relation between position- and momentum-space wavefunctions is

$$\langle p|\mathbb{1}|\psi\rangle = \langle p| \int_{-\infty}^{\infty} |x\rangle\langle x| \, \mathrm{d}x |\psi\rangle = \int_{-\infty}^{\infty} \langle p \,|\, x\rangle \psi(x) \, \mathrm{d}x = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}\psi(x) \, \mathrm{d}x.$$

This is a Fourier transform. The momentum space wavefunction is the fourier transform of the position-space wavefunction.

Lecture 18

Thu 03 Apr 2025 14:00

Recall: we identified momentum space eigenstates and eigenvalues $(p, |p\rangle)$, and we found

$$\langle p \mid x\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{-ipx/\hbar}.$$

Using this, we found the wavefunction $\langle p \mid \psi\rangle \coloneqq \widetilde{\psi}(p)$ in position space is given by the Fourier transform of $\psi(x)$:

$$\langle p \mid \psi\rangle = \int_{-\infty}^{\infty} \langle p \mid x\rangle\langle x \mid \psi\rangle \,\mathrm{d}x = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\hbar}} e^{-ipx/\hbar}\psi(x) \,\mathrm{d}x.$$

Similarly,

$$\psi(x) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}\widetilde{\psi}\,(p)\ \mathrm{d}p$$

is the reverse Fourier transform.

Now let's work with wavepackets. We never actually see position and momentum eigenstates, since we cannot do infinitely precise measurements, and since a particle cannot spread throughout the entire universe. A real particle is usually a wavepacket.

To minimize $\Delta p \Delta x$, and this is minimized by Gaussians. Recall that the time dependence is given by $\exp\left(\frac{ip(x - v^{(ph)}t}{\hbar}\right)$, for $v^{(ph)} = E_p/p = p/2m$.

Taking the Schrodinger equation with $V(x) = 0$, the eigenstates are momentum eigentates $|p\rangle$. We then set

$$|\psi, t\rangle = \int_{-\infty}^{\infty} f(p)|p\rangle e^{-iE_pt/\hbar} \,\mathrm{d}p.$$

We then have

$$\psi(x) = \int_{-\infty}^{\infty} f(p)\langle x \mid p\rangle e^{-iE_pt/\hbar} \,\mathrm{d}p = \int_{-\infty}^{\infty} f(p)e^{-ipx/\hbar}e^{-iE_pt/\hbar} \,\mathrm{d}p = \int_{-\infty}^{\infty} f(p)\exp\left(-\frac{ipx + iE_pt}{\hbar}\right) \mathrm{d}p.$$

Now we really need to identify $f(p)$, the coefficient function. Since we expect the particle to be a wavepacket, we expect $|f(p)|^2$ to appear to be a Gaussian. We Taylor expand E_p:

$$E_p = E_{p_0} + \frac{\mathrm{d}E_{p_0}}{\mathrm{d}p_0}(p - p_0) + \dots.$$

Plugging in,

$$\psi(x) = \int_{-\infty}^{\infty} f(p)\exp\left(-i\frac{(E_{p_0} + (p - p_0)E'_{p_0})t + ipx}{\hbar}\right) \mathrm{d}p.$$

Now we pull out p-independent terms.

$$\psi(x, t) = \exp\left(\frac{i(p_0E'_{p_0} - E_{p_0}t}{\hbar}\right)\int_{-\infty}^{\infty}\exp\left(\frac{ip(x - E'_{p_0}t}{\hbar}\right) f(p)\,\mathrm{d}p.$$

The front is a phase factor, so we can ignore it for now. The exponential inside the integral looks a lot like the Fourier transform of f:

$$\widetilde{f}(x) \propto \int_{-\infty}^{\infty} f(p)e^{ipx} \,\mathrm{d}p.$$

We can do the following substitution:

$$\widetilde{f}\left(\frac{x - E'_{p_0}t}{\hbar}\right) \propto \int_{-\infty}^{\infty} f(p)\exp\left(ipt\frac{x - E'_{p_0}}{\hbar}\right) \mathrm{d}p.$$

Therefore,

$$\psi(x,t) = \varphi \widetilde{f}\left(\frac{x - E_p' t}{\hbar}\right)$$

for φ the phase factor. We find that the true velocity $v^{(g)}$ (known as the group velocity) is given by

$$\frac{\mathrm{d}E_p}{\mathrm{d}p} = \frac{p}{m}.$$

We substitute in $p = p_0$ for the initial state here.

Lecture 19: Scawattering

Tue 08 Apr 2025 13:59

We will start with fixed-target scattering. A beam of particles goes towards some target and then it goes off in some direction. We try to predict stuff like what angle the particles bounce in. We will simplify it to 1 dimension. Suppose a beam of particles is zooming towards a target. The beam can now either go through the target, or bounce off of it. Our goal is to predict the result: how much goes forward (transmitted across), and how much goes backwards (reflected). This is summarized by the transmission coefficient T: the fraction of particles transmitted across, and R: the fraction of particles reflected.

Consider a finite potential well $V = V_0$, and consider a hamiltonian H. Recall we can shift all energies without modifying the system, so consider $H - V_0 \mathbb{1}$ for some constant V_0. This makes the energies at $\pm\infty$ go to 0, which is nice. This makes the potential nonzero in the well however; it becomes $-V_0$. For the region inside the well, we have

$$\left(\frac{-\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} - V_0\right)\psi(x) = E\psi(x).$$

It follows that

$$\frac{-\hbar^2}{2m}\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} = (E + V_0)\psi(x)$$
$$\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} = -\frac{2m(E+V_0)}{\hbar^2}\psi(x).$$

Setting this equal to $k^2\psi(x)$, the solutions are $\exp(\pm ikx)$.

For the regions outside the well, we have our solutions $k^2\exp(\pm i\ell x)$, where

$$\ell^2 = \frac{2mE}{\hbar^2}.$$

We thus have the solutions

$$\psi(x) = Ae^{ikx} + Be^{-ikx}, \qquad x \in (-a, a).$$
$$\psi(x) = A'e^{i\ell x} + B'e^{-i\ell x}, \qquad x > a.$$
$$\psi = A''e^{i\ell x} + B''e^{-i\ell x}, \qquad x < -a.$$

Let's return to our original question. We have electrons being shot at the potential, and they can either be transmitted or reflected. There is a particle gun on the left. That means particles are all moving in the $+x$ direction. This removes the $B'e^{-i\ell x}$ term for $x > 0$, since no particules come from that direction. This is fine since these are complex exponentials, so their magnitude is always 1.

The bigger our A'' coefficient is, the more particles we have coming in. The bigger A' is, the more particles we have going out. So we define

$$T \coloneqq \left|\frac{A'}{A''}\right|^2.$$

This is the probability of a particle going through the target, calculated with Born's rule. Similarly, the probability of reflection is

$$R \coloneqq \left|\frac{B''}{A''}\right|^2.$$

This is because B'' is the particles traveling back towards the shooter, meaning they've been reflected.

What happens inside the potential well doesn't matter, because our variables are now dependent on A'', which we control - its the amount of particles being shot.

We need to show that $T + R = 1$, to maintain the fact that these are in fact probabilities.

Intereting aside: this whole class is a nonrelativistic approximation where $c = \infty$ and the number of particles is conserved.

Define the probability current

$$j(x,t) = \frac{\hbar}{2im}\left(\psi^*(x,t)\left(\frac{\partial}{\partial x}\psi(x,t)\right) - \left(\frac{\partial}{\partial x}\psi^*(x,t)\right)\psi(x,t)\right).$$

We will show this satisfies a conservation condition

$$\frac{\partial}{\partial t}\left|\psi(x,t)\right|^2 = \frac{\partial}{\partial x}j(x,t).$$

To prove this, note that

$$\frac{\partial}{\partial x}j(x,t) = \frac{\hbar}{2im}\left(\frac{\partial\psi^*}{\partial x}\frac{\partial\psi}{\partial x} + \psi^*(x)\frac{\partial^2\psi}{\partial x^2} - \frac{\partial^2\psi^*}{\partial x^2}\psi(x) - \frac{\partial\psi^*}{\partial x}\frac{\partial\psi}{\partial x}\right).$$

Simplifying, we have

$$\frac{\partial j}{\partial x} = \frac{\hbar}{2im}\left(\psi^*(x,t)\frac{\partial^2\psi}{\partial x^2} - \frac{\partial^2\psi^*}{\partial x^2}\psi(x,t)\right).$$

The schrodinger equation says

$$i\hbar\frac{\mathrm{d}}{\mathrm{d}t}\psi(x,t) = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2}.$$

Some manipulation of this gives

$$\frac{\partial^2\psi}{\partial x^2} = \frac{2m}{\hbar^2}i\hbar\frac{\partial\psi}{\partial t}.$$

Taking the complex conjugate gives something like

$$\frac{\partial^2\psi^*}{\partial x^2} = \frac{2m}{-i\hbar}\frac{\partial\psi^*}{\partial t}.$$

Plugging this into our previous equation, we do some painful simplifications and arrive at our relation. fucking friend removed me on social media as a *joke* and i totally zoned out for the rest of the lecture dealing with that now i have to fucking learn this on my own kms

Lecture 20: Finishing Scattering

Tue 15 Apr 2025 14:00

Ok recall the scenario. We have a potential well, and we set the boundaries $V \to \pm\infty$ to 0, and the potential becomes $-V_0$ for $x \in (-a, a)$. The most general solution to Schrodinger's equation in this case is

$$\psi(x) = Ae^{ik_1x} + Be^{-ik_1x}, \qquad x < -a,$$

$$\psi(x) = Ce^{ik_2x} + De^{-ik_2x}, \qquad -a < x < a,$$

$$\psi(x) = Fe^{ik_1x}.$$

The final term goes away because particles that make it past the well HAVE to be going in the opposite direction of our particle shooter. We also have

$$k_1^2 = \frac{2m}{\hbar^2}E, \qquad k_2^2 = \frac{2m}{\hbar^2}\left(E + V_0\right).$$

This all follows from the Schrodinger equation. Our boundary conditions will be that the solutions must match across the boundary, and the derivative must also match since the derivative is continuous. That is, our boundary conditions are that ψ, ψ' are continuous. Computing the solution in the first region,

$$\psi(-a) = Ae^{-ik_1a} + Be^{ik_1a}.$$

In the second region,

$$\psi(-a) = Ce^{-ik_2a} + De^{ik_2a}.$$

These must be equal. We also have

$$Ce^{ik_2a} + De^{-ik_2a} = Fe^{ik_1a}$$

by continuity at a. We also have these same conditions with the derivatives:

$$ik_2Ce^{ik_2a} - ik_2De^{-ik_2a} = ik_1Fe^{ik_1a},$$

$$ik_1Ae^{-ik_1a} - ik_1Be^{ik_1a} = ik_2Ce^{-ik_2a} - ik_2De^{ik_2a}.$$

Since A corresponds to particles coming out of the gun, we can choose A. This reduces this to a system of 4 equations with 4 variables, which is solvable. We also define

$$T \coloneqq \left|\frac{F}{A}\right|^2, \qquad R = \left|\frac{B}{A}\right|^2.$$

These are the transmission and reflection coefficients, respectively. These are the probability that a particle is transmitted (goes out in F direction) or reflected (returns along B term), divided by A (magnitude of particles shot). We have that $T + R \equiv 1$ due to probabilities.

If we plug this into mathematica (or do a lot of manual computation), we obtain

$$T = \frac{1}{1+\xi}, \qquad R = \frac{1}{1+\xi^{-1}}, \qquad \xi \coloneqq \frac{\left(k_1^2 - k_2^2\right)^2 \sin^2(2k_2a)}{4k_1^2k_2^2}.$$

If k_2 is $\pi n/a$ for $n \in \mathbb{Z}$, then ξ goes to 0, and $T \to 1$. So if $k_2 = \pi n/a$, then all particles are transmitted.

Now let's do a potential bump. Our potential changes to be

$$V(x) = V_0, \ -a < x < a.$$

In essence, the new $V(x)$ is the negative of the old potential. We can reuse our old solutions and just flip a sign in k_2^2 to reflect this:

$$k_2^2 = \frac{2m}{\hbar}\left(E - V_0\right).$$

However, if V_0 becomes bigger than E, then k_2 becomes imaginary. Let

$$q^2 \coloneqq \frac{2m}{\hbar}\left(V_0 - E\right),$$

which is real. Moreover, $q = i^2k_2$ when k_2 is real, so we simply have

$$\psi(x) = Ce^{qx} + De^{-qx}.$$

Showing this is not difficult, it just amounts to solving the Schrodinger equation. The new ξ is just

$$\xi = \frac{\left(k_1^2 - (iq)^2\right)^2 \sin^2\left(2(iq)a\right)}{4k_1^2 (iq)^2} = \frac{\left(k_1^2 + q^2\right)^2 \sin^2\left(2iqa\right)}{-4k_1^2 q^2} = \frac{\left(k_1^2 + q^2\right)^2 \sinh^2\left(2qa\right)}{4k_1^2 q^2}.$$

Note here that we used the identity

$$\sin(ix) = \frac{e^{-i(ix)} - e^{i(ix)}}{2i} = \frac{e^{-x} - e^{x}}{2i} = i\sinh(x).$$

Traversing the bump then amounts to quantum tunneling. The fact that the decay inside the bump is exponetial, we can make very precise measurement tools that are very sensitive to the distance the particles have to tunnel over.

Chapter 5

Three Dimensional Systems

We now have wavefunctions $\psi(x,y,z)$. To define this, let $X|x,y,z\rangle = x|x,y,z\rangle$, $Y|x,y,z\rangle = y|x,y,z\rangle$, and $Z|x,y,z\rangle = z|x,y,z\rangle$. These are position operators in the x,y,z directions. We usually write $\mathbf{r}$ for (x,y,z), and write $\mathbf{R} = (X,Y,Z)$. We write r_i or R_i for the ith component. We then define

$$\psi(x,y,z) \coloneqq \langle x,y,z \mid \psi\rangle.$$

More simply,

$$\psi(r) = \langle r \mid \psi\rangle.$$

Oh yeah I'm too lazy to keep dragging the mathbf around. The old hamiltonian was

$$H = \frac{P^2}{2m} + V(X).$$

Our new one is then just

$$H = \frac{P^2}{2m} + V(R),$$

where

$$\mathbf{P} = i\hbar\nabla.$$

Issue: how do we square vectors? When we write $\mathbf{v}^2$, we mean $\mathbf{v}\cdot\mathbf{v}$. So $\mathbf{P}^2 = -\hbar^2\nabla\cdot\nabla = -\hbar^2\Delta$, for Δ the Lapalacian. I actually think I prefer the notation ∇^2. Our new hamiltonian is this

$$H = \frac{-\hbar^2}{2m}\nabla^2 + V(R).$$

This preserves our 1-dimensional magnitudes

$$P_j = -i\hbar\frac{\partial}{\partial j}.$$

The commutators all still have $[R_i, P_j] = i\hbar$. The energy eigenvalue equation becomes

$$\left(-\frac{\hbar^2}{2m}\nabla^2 + V(R)\right)\psi(r) = E\psi(r).$$

Because this is a PDE, it's basically unsolvable for most potentials $V(r)$. However, we *can* solve it for the *hydrogen atom*.

The hydrogen atom has various symmetries. The hydrogen atom has one proton and one electron. We thus have a wavefunction $\psi(r_e, r_p)$ which is 6-dimensional wtf that's terrifying. What is the potential $V(r_e, r_p)$? They have a Coulomb force between them, which is given by

$$V = -(4\pi\varepsilon_0)\frac{e^2}{|r_e - r_p|}.$$

The potential is translation invariant, as it only depends on relative distance. I think we will be able to WLOG set $r_p = 0$, and set it to a 3d function. We will see next class.

Lecture 21: Hydrogen Atom Work

Thu 17 Apr 2025 14:02

Recall we are working with the hydrogen atom, with an electron r_e and a proton r_p. The Coulomb potential is translation invariant

$$V = -\frac{e^2}{4\pi\varepsilon_0 \left|r_e - r_p\right|}.$$

That is, moving the *entire wavefunction* somewhere else preserves physics. The Schrodinger equation is given in the sum of the energies

$$H = \frac{P_p^2}{2m_p} + \frac{P_e^2}{2m_e} + V(|R_e - R_p|).$$

We can do a change of basis to write $\psi(r_e, r_p)$ in 3 coordinates. We write ψ in terms of the relative position $R_{\text{rel}} = R_e - R_p$. However, since R_{rel} only has 3 variables, we need 3 more for ψ. We use the center of mass R_{cm}:

$$R_{\text{cm}} = \frac{m_e R_e + m_p R_p}{m_e + m_p}.$$

Remark: $|r_e, r_p\rangle$ is indeed a tensor product $|r_e\rangle \otimes |r_p\rangle$, which is necessary since we are working in 2 particles. So P_p is like $1 \otimes P$, and vice versa for P_e.

We should be able to write H as $H_{\text{cm}} + H_{\text{rel}}$, since we should be able to write it as a hamiltonian for the entire atom (center of mass), and a hamiltonian for the internal structure (relative position).

We have $P_{\text{cm}} = P_e + P_p$, which can be verified easily by showing $[X_{\text{cm}}, P_{\text{cm}}] = i\hbar$ in this definition, where

$$P_{x,\text{cm}} = -i\hbar \left(\frac{\partial}{\partial x_e} + \frac{\partial}{\partial x_p} \right).$$

The relative momentum is given by

$$P_{\text{rel}} = \frac{m_e P_e - m_p P_p}{m_e + m_p}.$$

This gives a change of variables $\nabla_e, \nabla_p \to \nabla_{\text{cm}}, \nabla_{\text{rel}}$. This definition of P_{rel} ensures this satisfies the chain rule; in that $P_{\text{rel}} = -i\hbar\nabla_{\text{rel}}$.

Returning to our original hamiltonian

$$H = \frac{P_p^2}{2m_p} + \frac{P_e^2}{2m_e} + V\left(R_{\text{rel}}\right),$$

this turns into

$$H = \frac{P_{\text{cm}}^2}{2M} + \frac{P_{\text{rel}}^2}{2\mu} + V\left(R_{\text{rel}}\right),$$

where the reduced mass μ is

$$\frac{1}{\frac{1}{m_e} + \frac{1}{m_p}}.$$

This can be brute forced by plugging in the substitution. The hamiltonian now becomes two parts, the center of mass hamiltonian $P_{\text{cm}}^2/2M$, and the relative motion part which is the rest. The hamiltonians commute, so they can be diagonalized independently. WLOG, we can thus write $\psi(r_{\text{rel}}, r_{\text{cm}})$, which can be further written as $\psi_{\text{rel}}(r_{\text{rel}}) \otimes \psi_{\text{cm}}(r_{\text{cm}})$. The tensor product then collapses to

$$H|\psi\rangle = \left(1 \otimes H_{\text{cm}} + H_{\text{rel}} \otimes 1\right)\right) \left(|\psi_{\text{cm}}\rangle \otimes |\psi_{\text{rel}}\rangle\right) = \left(E_{\text{cm}} + E_{\text{rel}}\right) \left(|\psi_{\text{cm}}\rangle \otimes |\psi_{\text{rel}}\rangle\right).$$

Solving the first equation

$$H_{\text{cm}}|\psi_{\text{cm}}\rangle = E_{\text{cm}}|\psi_{\text{cm}}\rangle,$$

we write

$$\left(\frac{-\hbar^2}{2m}\nabla^2_{\text{cm}}\right)\psi_{\text{cm}}(r_{\text{cm}}) = E_{\text{cm}}\psi_{\text{cm}}(r_{\text{cm}}).$$

If we assume separability, this is solved by exponentials. This can be shown to be solved by

$$\frac{1}{\left(\sqrt{2\pi\hbar}\right)^3}\exp\left(\frac{i}{\hbar}\left(p_x x + p_y y + p_z z\right)\right) = \frac{1}{\left(\sqrt{2\pi\hbar}\right)^3}\exp\left(\frac{i\mathbf{p}\cdot\mathbf{r}}{\hbar}\right),$$

where the solutions to the separable equations are

$$\frac{1}{\sqrt{2\pi\hbar}}e^{ip_{x_i}x_i}.$$

We've solved $|\psi_{\text{cm}}\rangle$, and now we have to real with the relative motion. Let

$$H_{\text{rel}}|\psi_{\text{rel}}\rangle = E_{\text{rel}}|\psi_{\text{rel}}\rangle.$$

In position space, this looks like

$$\left(\frac{\hbar^2}{2\mu}\nabla^2_{\text{rel}} + V(R_{\text{rel}})\right)\psi_{\text{rel}}(r_{\text{rel}}) = E_{\text{rel}}\psi_{\text{rel}}(r_{rel}).$$

We will drop the "rel" subscripts, since this is the final equation we will solve this semester.

To solve this, we need to use spherical harmonics, since there is also going to be spherical symmetry. We need to take (x, y, z) and map it to (r, θ, ϕ). The change of variables is

$$x = r\sin\theta\cos\phi,$$

$$y = r\sin\theta\sin\phi,$$

$$z = r\cos\theta.$$

This will allow us to separate it into a solution depending on r, and a solution depending on (θ, ϕ), allowing it to be solved.

First: what is ∇^2 in spherical coordinates? We have

$$\nabla^2 = \frac{1}{r^2}\left(\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right).$$

Let's define

$$\hat{\nabla}^2 \coloneqq \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}.$$

This is called the Laplacian on the sphere. For the θ, ϕ part there is no potential, so it may be more simple than the r part. Our goal is to find eigenfunctions $Y(\theta, \phi)$ of $\hat{\nabla}^2$, and we can just factor in the $1/r^2$ later. These eigenfunctions are called the *spherical harmonics*.

Lecture 22: Spherical Harmonics Day 1

Tue 22 Apr 2025 14:03

Recall the Laplacian on the sphere:

$$\hat{\nabla}^2 = \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}.$$

We want functions $Y(\theta, \varphi)$ which are eigenfunctions of $\hat{\nabla}^2$. In this case, φ cycles through 2π and θ is the angle that runs through π. Let

$$\hat{\nabla}^2 Y(\theta, \varphi) = \lambda f(\theta, \varphi).$$

Our Schrodinger equation is

$$H\psi(r, \theta, \varphi) = \left(\frac{P^2}{2\mu} + V(r)\right)\psi(r, \theta, \varphi_= \left(-\frac{\hbar^2}{2\mu}\nabla^2 - \frac{e^2}{4\pi\varepsilon_0 r}\right)\psi(r, \theta, \varphi).$$

We set

$$\psi(r, \theta, \varphi) = \sum_i R_i(r) Y_i(\theta, \varphi).$$

Note that

$$\nabla^2 = \frac{1}{r^2}\left(\frac{\partial}{\partial r}\left(r^2 \frac{\partial}{\partial r}\right) + \hat{\nabla}^2\right).$$

We then have by eigenfunctions that

$$\left(-\frac{\hbar^2}{2\mu}\frac{1}{r^2}\left(\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \lambda\right)\right) R_i(r) Y_i(\theta, \varphi) = E R_i(r) Y_i(\theta, \varphi).$$

We can divide out by $Y_i(\theta, \varphi)$ to get an equation for the radial piece. This is why we need to solve for the spherical harmonics. After that, solving the radial equation gives us the eigenvalues. We sometimes use two indicies for these, and denote them $Y_{\ell m}$. We will find that

$$Y_{\ell m} = A_{\ell m}(-1)^m P_\ell^{|m|}(\cos\theta) e^{im\varphi}$$

for A a normalization constant and P an Associated Legendre Polynomial; meaning $P(\cos\theta)$ denotes a polynomial in $\cos\theta$. We also require $\ell \in \mathbb{Z}_{\geq 0}$ and m an integer such that $-\ell \leq m \leq \ell$.

Our goal right now is to solve

$$\left(\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right) Y_{\ell m}(\theta, \varphi) = \lambda Y_{\ell m}(\theta, \varphi).$$

We use separation of variables to write $Y(\theta, \varphi) = \Theta(\theta)\Phi(\varphi)$. We can use the same trick to get rid of the $\Theta(\theta)$ piece. We set

$$\frac{\partial^2}{\partial\varphi}\Phi(\varphi) = -\eta\Phi(\varphi).$$

If we can solve this, we would have

$$\left(\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) - \frac{1}{\sin^2\theta}\eta\right)\Theta(\theta)\Phi(\varphi) = \lambda\Theta(\theta)\Phi(\varphi).$$

This is trivial, we have

$$\Phi(\varphi) = e^{\pm i\varphi\sqrt{\eta}}.$$

The i makes this a function of the sphere. This means that we can impose the condition that $\Phi(\varphi) = \Phi(\varphi + 2\pi)$, constraining what η can be:

$$\exp\left(\pm i\varphi\sqrt{\eta} \pm i2\pi\sqrt{\eta}\right) = \exp\left(\pm i\varphi\sqrt{\eta}\right).$$

We have

$$\exp\left(\pm i2\pi\sqrt{\eta}\right) = 1 \implies \sqrt{\eta} \in \mathbb{Z}.$$

Therefore,

$$\eta = m^2, m \in \mathbb{Z}.$$

We now have eigenvalue/function pairs $(\eta_m, \Phi_m(\varphi)) = (m^2, e^{\mathrm{i}m\varphi})$. The $-$ sign gets included by letting m be negative.

Now we have
$$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)\Theta(\theta)\Phi(\varphi) - \frac{\eta}{\sin^2\theta}\Theta(\theta)\Phi(\varphi) = \lambda\Theta(\theta)\Phi(\varphi).$$
Cancelling,
$$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)\Theta(\theta) - \frac{\eta}{\sin^2\theta}\boldsymbol{\Theta}(\theta) = \lambda\Theta(\theta).$$
The solution has to be sines and cosines because it always is. We thus have two cases: whether $\boldsymbol{\Theta}$ is even or odd. We guess Θ is a polynomial in sin and cos, meaning each term looks like $\sin^n\theta\cos^m\theta$. If Θ is even, we must have n even, and substitute in $2n$:
$$\sin^{2n}\theta\cos^m\theta = \left(\sin^2\theta\right)^n\cos^m\theta = \left(1-\cos^2\theta\right)^n\cos^m\theta.$$
We can thus take even cases fully in terms of cosines. We then have that $\Theta(\theta) = a_j(\cos\theta)^j$ in Einstein notation. Plugging this into the eigenproblem is tedious but possible at this stage, and gives us
$$\sum_{j=1}^{\ell} a_j\left(j(j-1)\cos^j\theta - (2j^2+m^2+\lambda)\cos^{2+j}\theta + (j(j+1)+\lambda)\cos^{4+j}\theta\right) = 0.$$
The highest order term in the sum is
$$(\ell(\ell+1)+\lambda)\cos^{\ell+4}\theta.$$
This means that none of the cosines can cancel this term. This term thus *has* to go to 0, and thus
$$\lambda = -\ell(\ell+1).$$
We must have the coefficients of each linearly independent term vanish, so for any $j+4 \le \ell$, we end up with a relation between the coefficients:
$$a_j(j+4)(j+3) - a_{j-2}(2(j+2)^2 + m^2 - \ell(\ell+1)) + a_{j-4}(j(j+1) - \ell(\ell+1)) = 0.$$
This gives us an inductive relation. We can choose $a_\ell = 1$ to set the normalization, and choose all a_j with $j > \ell$ equal to 0. We inductively derive all other a_js. It can be shown that $a_{-j} = 0$ if ℓ is non-negative. We will show this when we do the *algebraic* approach tomorrow. Normalizing gives
$$\langle Y_1 \mid Y_2\rangle = \int_0^{2\pi}\int_0^{\pi} Y_1^* Y_2 \sin\theta\,\mathrm{d}\theta\mathrm{d}\varphi.$$
To compute $A_{\ell m}$, we demand that $\langle Y_1 \mid Y_2\rangle \equiv 1$.

Lecture 23: Spherical Harmonics: Algebraic Approach

Thu 24 Apr 2025 14:02

For a fixed ℓ, a rotation $Y_{\ell m}$ becomes a linear combination of valid $Y_{\ell m}$s, with ℓ fixed and m varying. This means that each $Y_{\ell m}$ induces a representation of the rotation group.

We are concerned with the angular momentum operators $\mathbf{L} = \mathbf{R}\times\mathbf{P}$, with $\mathbf{R}$ the tensor of position operators and $\mathbf{P}$ the tensor of momentum operators. Explicitly,
$$L_x = YP_z - ZP_y, \qquad L_y = ZP_x - XP_z, \qquad L_z = XP_y - YP_x.$$
An example of an explicit representation in position space is
$$L_x = -i\hbar y\frac{\partial}{\partial z} + i\hbar z\frac{\partial}{\partial y}.$$

Recall that the commutators of the angular momentum operators are

$$[P_i, P_j] = i\hbar\delta_{ij}.$$

To show this, recall that

$$[R_k, P_k] = i\hbar$$

for R_k a position space operator, and

$$[R_j, R_k] = 0, \qquad [R_j, P_k] = \delta_{jk} i\hbar, \qquad [P_j, P_k] = 0.$$

We do not necessarily require $j \neq k$ here. To compute angular momenta commutators, we then have

$$\begin{aligned}
[L_x, L_y] &= [YP_z - ZP_y, ZP_x - XP_z] \\
&= [YP_z, ZP_x] - [ZP_y, ZP_x] - [YP_z, XP_z] + [ZP_y, XP_z] \\
&= [YP_z, ZP_x] + [ZP_y, XP_z] \\
&= Y\,[P_z, Z]\,P_x + X\,[Z, P_z]\,P_y \\
&= -i\hbar Y P_x + i\hbar X P_y \\
&= i\hbar L_z.
\end{aligned}$$

In the third step, we removed commutators where all operators commute with each other. The fourth step follows from $[A, BC] = B\,[A, C] + [A, B]\,C$. The other commutators are calcuated in the same. This is called the *algebra* of 3d rotations $\mathfrak{so}(3)$. This is the same algebra as the spin operators, because they describe intrinsic angular momentum of particles.

Recall that $\hat{\nabla}^2$ had eigenvalues $\ell(\ell+1)$. To compute this, we need to find a rotational invariant. Given a vector $\mathbf{r}$, the inner product $\mathbf{r}\cdot\mathbf{r}$ is invariant under rotation operators. It follows that $\mathbf{L}\cdot\mathbf{L} = \mathbf{L}^2$ is invariant under action by SO(3). Also note that the commutator $[A, L_x]$ with any operator A should be viewed as a rotation around the x-axis. By this logic,

$$\left[\mathbf{L}^2, L_x\right] = 0.$$

This holds for any L_i obviously. We can then assume WLOG the can simultaneously diagonalize $\mathbf{L}^2$ and any L_i.

What is $\mathbf{L}^2$ in position space? Since it is rotationally invariant, and since each term looks like double derivatives if we expand it out, we can actually show (fairly easily but tediously) that

$$\mathbf{L}^2 = \hbar^2\hat{\nabla}^2.$$

Let's show it. First, we go to spherical coordinates. We have

$$z = \cos\theta, \qquad x = r\sin\theta\cos\varphi, \qquad y = r\sin\theta\sin\varphi.$$

The derivative (wrt x) is

$$\frac{\partial}{\partial x} = \sin\theta\cos\varphi\frac{\partial}{\partial r} + \frac{\cos\theta\cos\varphi}{r}\frac{\partial}{\partial\theta} - \frac{\sin\varphi}{r sin\theta}\frac{\partial}{\partial\varphi}.$$

Plugging in and doing a lot of algebra, we have that

$$L_z = -i\hbar\frac{\partial}{\partial\varphi}.$$

It's easier to compute L_x and L_y as $L_\pm := L_x \pm iL_y$. Doing a lot of tedious algebra, we obtain

$$L_\pm = \hbar e^{\pm i\varphi}\left(\pm\frac{\partial}{\partial\theta} + i\cos\theta\frac{\partial}{\partial\varphi}\right).$$

We then have
$$L_+L_- = (L_x + iL_y)(L_x - iL_y) = L_x^2 + L_y^2 + i[L_y, L_z].$$
Since the commutator is $-i\hbar L_z$, the right term simplifies to $\hbar L_z$. We then have
$$\mathbf{L}^2 = L_x^2 + L_y^2 + L_z^2 = L_+L_- - \hbar L_z + L_z^2.$$
Plugging in gives us
$$\mathbf{L}^2 = -\hbar^2\left(\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right) = -\hbar^2\hat{\nabla}^2.$$

How can we find eigenstates of $\mathbf{L}^2$? Let
$$\mathbf{L}|\psi\rangle = -\hbar^2\lambda|\psi\rangle.$$
Since $\mathbf{L}^2$ and L_z commute, we take $|\psi\rangle$ to be an eigenstate of L_z. Let $L_z|\psi\rangle = m\hbar|\psi\rangle$, and rename the state $|\psi\rangle$ to $|\ell, m\rangle$.

Consider the commutator
$$[L_+, L_-] = [L_x + iL_y, L_x - iL_y] = -2i[L_x, L_y] = 2\hbar L_z.$$
We also ahve
$$[L_z, L_\pm] = [L_z, L_x] + i[L_z, L_y] = i\hbar L_y - i^2\hbar L_x = \pm\hbar L_\pm.$$
Finally,
$$\left[\mathbf{L}^2, L_\pm\right] = 0.$$

Note that in the second commutator, we have a relation that looks like $[A, B] = \lambda B$. This implies that B changes the A eigenvalue by λ:
$$A|a\rangle = a|a\rangle \implies AB|a\rangle = ([A, B] + BA)|a\rangle = (\lambda B + Ba)|a\rangle = (a + \lambda)B|a\rangle.$$
In other words, $B|a\rangle$ has an eigenvalue $AB|a\rangle = (a + \lambda)B|a\rangle$.

This means that L_+ changes the eigenvalues of L_z by $\hbar$. So we have
$$L_z|\ell, m\rangle = m\hbar|\ell, m\rangle \implies L_zL_+|\ell, m\rangle = (m+1)\hbar|\ell, m\rangle.$$
Similarly for L_-. We can then say $L_+|\ell, m\rangle \propto |\ell, m+1\rangle$. We also somehow obtained that
$$\mathbf{L}^2|\ell, m\rangle = \hbar^2\ell(\ell+1)|\ell, m\rangle.$$
This means that L_+ and L_- can move up and down in m without changing the overall eigenvalue of $\mathbf{L}^2$. This means that
$$L_+Y_{\ell m} = Y_{\ell, m+1}.$$
Additionally, we can then show that hitting a lowest-weight or highest-weight state gives you 0.

Lecture 24: Angular Momentum and Spherical Harmonics

Tue 29 Apr 2025 14:00

Recall we represented
$$L_z = -i\hbar\frac{\partial}{\partial\varphi}.$$
We will show that
$$e^{i\alpha L_z/\hbar}$$

is a rotation. Consider

$$e^{i\alpha L_z/\hbar} f(\varphi) = \sum_{n=0}^{\infty} \frac{(i\alpha L_z)^n}{\hbar^n n!} f(\varphi) = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\alpha \frac{\partial}{\partial\varphi}\right)^n f(\varphi) = \sum_{n=0}^{\infty} \frac{\alpha^n}{n!} \frac{\partial^n}{\partial\varphi^n} f(\varphi).$$

Note that the taylor series of $f(\varphi)$ at α, meaning $f(\varphi+\alpha)$, is precisely this series. This means we have increased φ, or we have rotated f by an angle α. Thus, L_z rotates things in a sense.

The important commutator relations we had are

$$\left[\mathbf{L}^2, L_\pm\right] = \left[\mathbf{L}^2, L_z\right] = 0, \qquad [L_z, L_\pm] = \pm\hbar L_\pm.$$

This means we can diagonalize $\mathbf{L}^2$ and L_z simultaneously, and we have

$$\mathbf{L}^2|\ell,m\rangle = \hbar^2\ell(\ell+1)|\ell,m\rangle, \qquad L_z|\ell,m\rangle = \hbar m|\ell,m\rangle.$$

The relation $\left[\mathbf{L}^2, L_\pm\right]$ tells us that given an $\mathbf{L}^2$-eigenvector, $L_\pm$ gives a new eigenvector with the same eigenvalue. The relation on the right tells us that $L_\pm$ acting on an L_z-eigenvector gives a new eigenvector of L_z with eval shifted by $\pm\hbar$.

This means

$$\begin{aligned} L_+|\ell,m\rangle &\propto |\ell,m+1\rangle, \\ L_-|\ell,m\rangle &\propto |\ell,m-1\rangle. \end{aligned}$$

Also recall that

$$L_+^\dagger = L_-,$$

and thus they are not hermitian.

Since eigenstates are taken to be normalized WLOG, consider

$$\left|L_+|\ell,m\rangle^2\right| = \langle\ell,m|L_+^\dagger L_+|\ell,m\rangle = \langle\ell,m|L_-L_+|\ell,m\rangle = \langle\ell,m|\left(\mathbf{L}^2 - L_z^2 - \hbar L_z\right)|\ell,m\rangle.$$

This last step follows from an identity

$$L_\mp L_\pm = \mathbf{L}^2 - L_z^2 \mp \hbar L_z.$$

Continuing:

$$\langle\ell,m|\left(\hbar^2\ell(\ell+1) - \hbar^2 m^2 - \hbar^2 m\right)|\ell,m\rangle = \hbar^2\left(\ell(\ell+1) - m(m+1)\right).$$

This must be nonnegative. We thus have $\ell \geq m$. Also note that when $\ell = m$, it goes to 0, so this is consistent with stepping with $L_\pm$. To compute spherical harmonics, we just use the minimum m. For example,

$$L_-Y_{2,-2} = 0 \implies -\hbar e^{-i\varphi}\left(\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\varphi}\right)Y_{2,-2} = 0.$$

We know the phi dependence is exponntial cuz eigenvectors so we get some trig crap

Lecture 25

Thu 01 May 2025 11:05